Combinatorial Privacy: Private Boolean Sums via Polyhedral Masking in Birkhoff Polytopes

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Abstract

We present PolyVeil, a protocol for computing aggregate counts of binary values held privately by multiple clients, and introduce *Combinatorial Privacy* as a new paradigm for privacy-preserving computation. A central server learns only the total number of ones across all client bitstreams, while individual client data remains protected. Our approach encodes private bits as permutation matrices, embeds them inside doubly stochastic matrices via convex combinations with random decoy permutations, and exploits the non-uniqueness of the Birkhoff–von Neumann decomposition to prevent the server from recovering any individual client's data. Unlike cryptographic paradigms (MPC, homomorphic encryption) that rely on computational hardness assumptions, or differential privacy that provides guarantees through noise addition, Combinatorial Privacy derives attack bounds from the provable non-uniqueness of decompositions in combinatorial structures — bounds that hold against computationally unbounded adversaries because they depend on combinatorial facts rather than hardness conjectures, though they do not constitute simulation-based indistinguishability. A secure shuffling step, scaling linearly in the number of clients, enables exact noise cancellation at the server without de-anonymization. We provide formal security analysis under the honest-but-curious model, including a $1/k$ attack bound for the compressed protocol and a $1/(k \cdot L)$ bound for the full protocol where L is the number of Birkhoff–von Neumann decompositions consistent with the server's observation. We derive precise values of L, showing it is at least $\Omega(n^2)$ for $K_t = 2$ decoy permutations per client, at least $\Omega(n^4)$ for $K_t = \Omega(n \log n)$, and exactly $(2n)!$ (factorial in the encoding dimension) when protocol parameters satisfy a specific interior condition. PolyVeil achieves exact answers with communication cost $O(k)$ in the number of clients while avoiding the computational overhead of fully homomorphic encryption and requiring only a single lightweight cryptographic primitive (a shuffler or summation functionality). We develop three protocol variants with progressively stronger security guarantees: a compressed variant achieving a $1/k$ attack bound with a shuffler, a full variant achieving $1/(k \cdot L)$ with BvN decomposition ambiguity, and an information-theoretically secure variant (PolyVeil-IT) that achieves *perfect simulation-based indistinguishability* — identical distributions for any two inputs with the same aggregate, with statistical distance zero — by replacing the shuffler with a summation functionality.

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1 Introduction

Computing aggregate statistics over private data held by many parties is a foundational problem in privacy-preserving computation. A concrete and widely applicable instance is the Boolean sum problem, in which k clients each hold a private binary string of length n and a server wishes to learn the total number of ones across all kn bits without learning any individual client's data. Applications range from epidemiological surveillance, where a health authority counts positive test results without accessing individual diagnoses, to federated analytics, where a service provider tallies binary feature occurrences without centralizing user data.

Existing approaches to this problem broadly fall into three categories. Generic secure multi-party computation (MPC) protocols, built on garbled circuits or secret sharing, provide strong composable security with guarantees based on specific computational hardness assumptions (such as the difficulty of integer factorization or the Learning with Errors problem over lattices), but impose communication and computation costs that scale with circuit complexity and become prohibitive when k is large or network conditions are constrained. Homomorphic encryption allows computation on ciphertexts, with security based on assumptions such as the composite residuosity problem (Paillier [5]) or the Learning with Errors problem (lattice-based FHE [6]), but carries substantial per-operation cost and requires careful key management. Differential privacy provides a framework for releasing aggregate statistics to untrusted parties by adding calibrated noise, offering formal (ε, δ)-guarantees that hold regardless of an adversary's computational power, but inherently sacrificing accuracy for privacy.

PolyVeil occupies a distinct point in this design space, one that we argue represents a new paradigm we call *Combinatorial Privacy*. In PolyVeil, the core idea is to encode each client's private bitstream as a permutation matrix, embed it inside a doubly stochastic matrix by mixing it with random decoy permutations, and use secure aggregation to recover the aggregate bit sum exactly. The Birkhoff–von Neumann theorem guarantees that every doubly stochastic matrix admits many decompositions into convex combinations of permutation matrices, and this non-uniqueness forms one of the two security layers of the protocol.

We present a rigorous security analysis that identifies a fatal vulnerability in naive implementations of this approach (the de-shuffling attack, which recovers all individual data with probability 1) and then develops a corrected *two-layer protocol* that achieves provable security. In the corrected protocol, the main server receives only aggregate scalars and is information-theoretically secure (its view is identically distributed for any two inputs with the same aggregate). A separate aggregator entity receives Birkhoff-encoded matrices but not the noise values, and faces a computational barrier: recovering the private permutation matrix from its encoding requires evaluating the density of a random doubly stochastic matrix at a given point, which we prove is #P-hard via a reduction to the permanent. The two layers compose so that no single entity can learn individual data: the server lacks the information, and the aggregator lacks the computational power.

This two-layer architecture distinguishes Combinatorial Privacy from existing paradigms. Unlike MPC and HE, which derive security from number-theoretic hardness (factoring, LWE), PolyVeil's computational layer derives from the #P-hardness of evaluating Birkhoff polytope decomposition likelihoods (proved for likelihood-based attacks; conjectured for all attacks). Unlike DP, PolyVeil produces exact answers. The protocol requires no public-key infrastructure and has $O(k)$ communication in the compressed variant.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 establishes mathematical preliminaries. Section 4 presents the protocol with correctness proofs and a worked example. Section 5 provides security analysis. Section 6 develops the geometric role of α^*. Section 7 compares with alternative approaches and analyzes attack resilience. Section 8 develops

the information-theoretically secure variant with a complete simulation proof. Section 9 concludes. Appendix A provides background on simulation-based proofs.

2 Related Work

Secure multi-party computation. The problem of computing functions over distributed private inputs has been studied since the foundational work of Yao [1] on garbled circuits and Goldreich, Micali, and Wigderson [2] on the GMW protocol. These generic constructions can compute any function securely, including Boolean sums, but their communication and computation costs scale with the circuit complexity of the target function. More recent frameworks such as SPDZ [3] reduce the online cost through preprocessing, but the per-gate overhead remains significant compared to the simple arithmetic in PolyVeil. Specialized secure aggregation protocols such as that of Bonawitz et al. [4] reduce communication through pairwise secret sharing and handle client dropout, achieving $O(k)$ per-client communication with $O(k^2)$ setup. PolyVeil achieves $O(1)$ per-client communication in its compressed variant without pairwise key agreement.

Homomorphic encryption. Additively homomorphic schemes such as Paillier [5] support additive aggregation natively. Each client encrypts their bit count under a common public key, the server multiplies ciphertexts, and a designated party decrypts the sum. This achieves exact results with IND-CPA security but requires public-key infrastructure. A Paillier ciphertext is typically 4096 bits at 128-bit security, whereas PolyVeil transmits a single scalar per client. Fully homomorphic encryption [6] generalizes to arbitrary computations but with substantially greater overhead.

Differential privacy. Differential privacy [8] provides formal privacy guarantees through calibrated noise. In the local model, each client randomizes their data before sending it to the server, achieving privacy without trust but with error $\Theta(\sqrt{kn}/\varepsilon)$. The central model achieves error $\Theta(n/\varepsilon)$ but requires a trusted curator to see raw data. The shuffle model [9, 10] interpolates by interposing an anonymous shuffler, achieving central-model accuracy with local-model trust. PolyVeil produces exact results and uses the same shuffling infrastructure, but derives privacy from algebraic masking rather than statistical noise.

Instance mixing and data obfuscation. InstaHide [11] mixes private data records with public datasets and random sign patterns for privacy-preserving machine learning. While both InstaHide and PolyVeil involve mixing private data with random elements, PolyVeil focuses on aggregation rather than prediction, achieves exact results through algebraic noise cancellation, and provides security guarantees rooted in the combinatorial structure of the Birkhoff polytope.

Secret sharing. Secret sharing schemes [13] distribute a secret among multiple parties so that only authorized subsets can reconstruct it. PolyVeil does not use secret sharing directly but instead exploits the structure of doubly stochastic matrices so that the private data is one of many valid decompositions of a publicly shared matrix, creating a computational barrier for any entity that observes the matrix but not the decomposition coefficients.

3 Preliminaries

3.1 Doubly Stochastic Matrices and the Birkhoff Polytope

Definition 3.1 (Doubly stochastic matrix)**.** *A square matrix $A = (a_{ij}) \in \mathbb{R}^{m\times m}$ with non-negative entries is* doubly stochastic *if every row and every column sums to one: $\sum_{j=1}^{m} a_{ij} = 1$ for all $i \in [m]$, and $\sum_{i=1}^{m} a_{ij} = 1$ for all $j \in [m]$.*

Definition 3.2 (Birkhoff polytope)**.** *The* Birkhoff polytope *$\mathcal{B}_m$ is the set of all $m \times m$ doubly stochastic matrices. It is a convex polytope in $\mathbb{R}^{m\times m}$ of dimension $(m-1)^2$.*

Theorem 3.3 (Birkhoff–von Neumann [21, 22])**.** *The vertices of $\mathcal{B}_m$ are precisely the $m \times m$ permutation matrices. Every doubly stochastic matrix $A \in \mathcal{B}_m$ can be written as a convex combination of permutation matrices:*

$$A = \sum_{i=1}^{r} \theta_i P_i, \qquad \theta_i > 0, \quad \sum_{i=1}^{r} \theta_i = 1,$$

where each P_i is a permutation matrix. This is a Birkhoff–von Neumann (BvN) decomposition *of A.*

BvN decompositions are generically *non-unique*: a doubly stochastic matrix in the interior of $\mathcal{B}_m$ admits many distinct decompositions.

Theorem 3.4 (Decomposition multiplicity, Brualdi [18])**.** *Let $A \in \mathcal{B}_m$ have p positive entries. The number of distinct BvN decompositions of A is at least p^2.*

Theorem 3.5 (Marcus–Ree [19])**.** *Every $m \times m$ doubly stochastic matrix can be expressed as a convex combination of at most $m^2 - 2m + 2$ permutation matrices. Finding a BvN decomposition with the minimum number of permutations is NP-hard [20].*

3.2 Uniform Sampling of Permutation Matrices

A $m \times m$ permutation matrix P corresponds bijectively to a permutation $\sigma \in S_m$ (the symmetric group on $[m] = \{1, \ldots, m\}$) via $P_{ij} = \mathbf{1}[\sigma(i) = j]$. That is, row i of P has its unique 1 in column $\sigma(i)$. To draw P *uniformly at random from* S_m, one draws a uniformly random permutation σ and constructs the corresponding matrix.

The standard algorithm for drawing a uniform random permutation is the Fisher–Yates shuffle (also known as the Knuth shuffle): starting from the identity permutation $\sigma = (1, 2, \ldots, m)$, for $i = m, m-1, \ldots, 2$, draw j uniformly at random from $\{1, \ldots, i\}$ and swap $\sigma(i) \leftrightarrow \sigma(j)$. This produces each of the $m!$ permutations with equal probability $1/m!$ and runs in $O(m)$ time using $O(m)$ random bits (specifically $\sum_{i=2}^{m} \lceil \log_2 i \rceil$ bits). For our protocol with $m = 2n$, each random permutation matrix costs $O(n)$ time and $O(n \log n)$ random bits.

3.3 Permutation Encoding of Binary Data

Definition 3.6 (Bit-to-permutation encoding)**.** *For a bit $b \in \{0, 1\}$, define the 2×2 permutation matrix:*

$$\Pi(b) = \begin{cases} I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} & \text{if } b = 0, \\ J_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} & \text{if } b = 1. \end{cases}$$

For a bitstream $\mathbf{b} = (b_1, \ldots, b_n) \in \{0,1\}^n$*, define the block-diagonal encoding:*

$$M(\mathbf{b}) = \mathrm{blockdiag}\big(\Pi(b_1),\ \Pi(b_2),\ \ldots,\ \Pi(b_n)\big) \in \{0,1\}^{2n\times 2n}.$$

Since each $\Pi(b_j)$ is a 2×2 permutation matrix, their block-diagonal assembly $M(\mathbf{b})$ is a $2n\times 2n$ permutation matrix and a vertex of $\mathcal{B}_{2n}$. The block-diagonal structure confines each bit to a disjoint 2×2 block, enabling the algebraic extraction we now develop.

Example 3.7 (Encoding of a 2-bit stream)**.** *Consider the bitstream* $\mathbf{b} = (1,0)$ *with* $n = 2$*. The encoding produces:*

$$M\big((1,0)\big) = \mathrm{blockdiag}(\Pi(1),\ \Pi(0)) = \mathrm{blockdiag}\left(\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix},\ \begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}\right) = \begin{pmatrix}0 & 1 & 0 & 0\\ 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 1\end{pmatrix}.$$

The first 2×2 *block (rows 1–2, columns 1–2) encodes* $b_1 = 1$ *as the swap matrix* J_2*. The second* 2×2 *block (rows 3–4, columns 3–4) encodes* $b_2 = 0$ *as the identity* I_2*. All entries outside these blocks are zero due to the block-diagonal structure.*

3.4 Algebraic Bit Count Extraction

Definition 3.8 (Selection vectors)**.** *Define* $\mathbf{w}, \mathbf{y} \in \mathbb{R}^{2n}$ *by:*

$$w_i = \begin{cases} 1 & \text{if } i \text{ is odd,}\\ 0 & \text{if } i \text{ is even,}\end{cases} \qquad y_i = \begin{cases} 0 & \text{if } i \text{ is odd,}\\ 1 & \text{if } i \text{ is even.}\end{cases}$$

Equivalently, $\mathbf{w} = (1,0,1,0,\ldots,1,0)^T$ *and* $\mathbf{y} = (0,1,0,1,\ldots,0,1)^T$*, each of length* $2n$*.*

Lemma 3.9 (Bit count extraction)**.** *For any bitstream* $\mathbf{b} = (b_1, \ldots, b_n) \in \{0,1\}^n$ *with permutation encoding* $M = M(\mathbf{b})$*:*

$$\mathbf{w}^T M\,\mathbf{y} \;=\; \sum_{j=1}^{n} b_j.$$

Proof. We compute $\mathbf{w}^T M\,\mathbf{y}$ by expanding the matrix-vector products step by step.

Computing $M\mathbf{y}$**.** Since $M = \mathrm{blockdiag}(\Pi(b_1), \ldots, \Pi(b_n))$ is block-diagonal, the product $M\mathbf{y}$ decomposes into independent block multiplications. For the j-th block, the relevant entries of $\mathbf{y}$ are $y_{2j-1} = 0$ and $y_{2j} = 1$. Writing $\mathbf{y}_j = (y_{2j-1},\ y_{2j})^T = (0,1)^T$ for the portion of $\mathbf{y}$ corresponding to block j, we have:

$$\Pi(b_j)\,\mathbf{y}_j = \Pi(b_j)\begin{pmatrix}0\\ 1\end{pmatrix}.$$

When $b_j = 0$: $\Pi(0)\binom{0}{1} = \begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}\binom{0}{1} = \binom{0}{1}$. When $b_j = 1$: $\Pi(1)\binom{0}{1} = \begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}\binom{0}{1} = \binom{1}{0}$. Therefore, the entries of $M\mathbf{y}$ at positions $2j-1$ and $2j$ are:

$$(M\mathbf{y})_{2j-1} = b_j, \qquad (M\mathbf{y})_{2j} = 1 - b_j. \tag{1}$$

To verify this: when $b_j = 0$, $(M\mathbf{y})_{2j-1} = 0 = b_j$ and $(M\mathbf{y})_{2j} = 1 = 1 - b_j$. When $b_j = 1$, $(M\mathbf{y})_{2j-1} = 1 = b_j$ and $(M\mathbf{y})_{2j} = 0 = 1 - b_j$. Both cases match (1).

Computing $\mathbf{w}^T(M\mathbf{y})$. Now we take the inner product of $\mathbf{w}$ with $M\mathbf{y}$:

$$\begin{aligned}
\mathbf{w}^T(M\mathbf{y}) &= \sum_{i=1}^{2n} w_i\,(M\mathbf{y})_i \\
&= \sum_{j=1}^{n} \Big[w_{2j-1}\,(M\mathbf{y})_{2j-1} \;+\; w_{2j}\,(M\mathbf{y})_{2j} \Big] \\
&= \sum_{j=1}^{n} \Big[1 \cdot b_j \;+\; 0 \cdot (1 - b_j) \Big] \\
&= \sum_{j=1}^{n} b_j, \qquad (2)
\end{aligned}$$

where the third equality substitutes $w_{2j-1} = 1$, $w_{2j} = 0$ from Definition 3.8 and $(M\mathbf{y})_{2j-1} = b_j$, $(M\mathbf{y})_{2j} = 1 - b_j$ from (1). The key mechanism is that $\mathbf{w}$ "selects" only the odd-indexed entries of $M\mathbf{y}$, each of which equals the corresponding bit b_j. □

Example 3.10 (Extraction for a concrete bitstream)**.** *Take $n = 2$, $\mathbf{b} = (1, 0)$. Then $\mathbf{w} = (1, 0, 1, 0)^T$ and $\mathbf{y} = (0, 1, 0, 1)^T$. From Example 3.7:*

$$M = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

First compute $M\mathbf{y}$:

$$M\mathbf{y} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \cdot 0 + 1 \cdot 1 + 0 \cdot 0 + 0 \cdot 1 \\ 1 \cdot 0 + 0 \cdot 1 + 0 \cdot 0 + 0 \cdot 1 \\ 0 \cdot 0 + 0 \cdot 1 + 1 \cdot 0 + 0 \cdot 1 \\ 0 \cdot 0 + 0 \cdot 1 + 0 \cdot 0 + 1 \cdot 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix}.$$

Checking against (1): for block $j = 1$ ($b_1 = 1$): $(M\mathbf{y})_1 = 1 = b_1$ and $(M\mathbf{y})_2 = 0 = 1 - b_1$. For block $j = 2$ ($b_2 = 0$): $(M\mathbf{y})_3 = 0 = b_2$ and $(M\mathbf{y})_4 = 1 = 1 - b_2$. Both match.

Now compute $\mathbf{w}^T(M\mathbf{y})$:

$$\mathbf{w}^T(M\mathbf{y}) = (1, 0, 1, 0) \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix} = 1 \cdot 1 + 0 \cdot 0 + 1 \cdot 0 + 0 \cdot 1 = 1.$$

The result is $1 = b_1 + b_2 = 1 + 0$, confirming the lemma.

4 The PolyVeil Protocol

4.1 Problem Statement

We consider k client entities, where client $t \in [k]$ holds a private binary bitstream $\mathbf{b}_t = (b_{t,1}, \ldots, b_{t,n}) \in \{0, 1\}^n$ of length n. Let $s_t = \sum_{j=1}^{n} b_{t,j}$ denote the number of ones in client t's bitstream. The goal is for a server to compute the aggregate $S = \sum_{t=1}^{k} s_t$ without learning any individual s_t, any individual bitstream $\mathbf{b}_t$, or any partial aggregates involving fewer than all k clients.

4.2 Threat Model

We operate in the *honest-but-curious* (semi-honest) model, defined as follows.

Server behavior. The server executes every instruction of the protocol exactly as specified. It does not deviate from the protocol by, for example, sending altered intermediate results to clients, injecting false data, or failing to perform a required computation. However, the server records every message it receives and may subsequently perform arbitrary polynomial-time computations on this recorded transcript in an attempt to infer individual client data. The server's computational power is bounded only by polynomial time: it may run brute-force searches over feasible spaces, solve optimization problems, and apply any statistical inference technique. The security guarantees we prove hold against any such polynomial-time analysis.

Communication security. All communication channels between each client and the server, and between each client and the shuffler, are authenticated and encrypted using standard transport-layer security (e.g., TLS 1.3). Authentication ensures that the server receives messages only from legitimate clients and not from impersonators. Encryption ensures that no external eavesdropper observing the network can read the content of any message. Together, these guarantees mean that the only entity that sees a client's message to the server is the server itself, and the only entity that sees a client's message to the shuffler is the shuffler itself. We do not assume that the communication channels hide metadata such as message timing or size.

Client behavior. Every client follows the protocol faithfully. No client modifies, omits, or fabricates any message. No client shares its private data, its random coins, or its intermediate computations with the server or with any other client (beyond what the protocol prescribes). In particular, no client colludes with the server to de-anonymize the shuffled values. The non-collusion assumption is essential: if even one client shared its η_t value directly with the server (outside the shuffle), the server could link that η_t to the client's identity and compute $s_t = (f_t - \eta_t)/\alpha^*$.

Trusted shuffler. There exists a functionality $\mathcal{F}_{\text{shuffle}}$ that operates as follows. It accepts as input one scalar value from each of the k clients, collecting the multiset $\{\eta_1, \ldots, \eta_k\}$. It then applies a permutation π drawn uniformly at random from the symmetric group S_k (the set of all $k!$ bijections on $[k]$), and outputs the permuted sequence $(\eta_{\pi(1)}, \ldots, \eta_{\pi(k)})$ to the server. Critically, the server learns the values in the output sequence but does *not* learn the permutation π. This means that for any position j in the output, the server knows the value $\eta_{\pi(j)}$ but cannot determine which client submitted it. The shuffler does not reveal π to the clients either.

This ideal functionality can be instantiated in several ways. The simplest is a *non-colluding auxiliary server*: a separate physical server, operated by an independent party that does not collude with the main server, receives all η_t values, permutes them, and forwards the result. A stronger instantiation is a *mixnet*, where each client encrypts its value under layered encryption addressed to a chain of relay servers, each of which peels one encryption layer and shuffles the messages. Verifiable mixnets [14] additionally produce a zero-knowledge proof that the output is a valid permutation of the input, preventing a malicious relay from altering values. A fully cryptographic instantiation uses a *multi-party shuffling protocol* such as the secret-shared shuffle of Chase, Ghosh, and Poburinnaya [14], which distributes the shuffling computation among two or more servers so that no single server learns the permutation, achieving security even if one server is corrupted.

We require $k \geq 3$ to ensure that the shuffle provides meaningful anonymity. With $k = 1$, the shuffled output contains a single value that is trivially linked to the sole client. With $k = 2$, the server has a 1/2 probability of guessing the correct assignment, providing negligible privacy.

4.3 Protocol Description

The protocol uses a public parameter $\alpha^* \in (0, 1)$ whose geometric role is analyzed in Section 6.

Algorithm 1 PolyVeil

Public parameters: bit length n, scaling factor $\alpha^* \in (0, 1)$, selection vectors $\mathbf{w}$, $\mathbf{y}$ (Definition 3.8).

<u>Client-Side Masking</u>

for each client $t \in \{1, \ldots, k\}$ in parallel **do**

  Encode private bitstream as $M_t = M(\mathbf{b}_t) \in \{0, 1\}^{2n \times 2n}$ via Definition 3.6.

  Choose $K_t \geq 2$ and draw K_t uniformly random permutation matrices $P_{t,1}, \ldots, P_{t,K_t}$ from S_{2n} using the Fisher–Yates shuffle (Section 3.2).

  Sample positive coefficients $\alpha_{t,1}, \ldots, \alpha_{t,K_t} > 0$ with $\sum_{i=1}^{K_t} \alpha_{t,i} = 1 - \alpha^*$.

  Construct the masked matrix:

$$D_t \;=\; \alpha^* \, M_t \;+\; \sum_{i=1}^{K_t} \alpha_{t,i} \, P_{t,i}. \tag{3}$$

  Compute the noise term:

$$\eta_t \;=\; \sum_{i=1}^{K_t} \alpha_{t,i} \, \big(\mathbf{w}^T P_{t,i} \, \mathbf{y}\big). \tag{4}$$

  Send D_t to the server.

end for

<u>Server-Side Extraction</u>

for each client $t \in \{1, \ldots, k\}$ **do**

  Server computes:

$$f_t \;=\; \mathbf{w}^T D_t \, \mathbf{y}. \tag{5}$$

end for

<u>Secure Noise Transmission</u>

Each client t submits scalar η_t to the trusted shuffler $\mathcal{F}_{\text{shuffle}}$.

Shuffler draws $\pi \xleftarrow{\$} S_k$ and sends $(\eta_{\pi(1)}, \ldots, \eta_{\pi(k)})$ to server.

<u>Aggregation</u>

Server computes:

$$S \;=\; \frac{1}{\alpha^*} \left(\sum_{t=1}^{k} f_t \;-\; \sum_{t=1}^{k} \eta_t \right). \tag{6}$$

4.4 Proof of Correctness

Theorem 4.1 (Correctness)**.** *Algorithm 1 computes $S = \sum_{t=1}^{k} s_t$ exactly.*

Proof. The matrix $M_t = M(\mathbf{b}_t)$ is a permutation matrix (Definition 3.6) and hence doubly stochastic: each row and column contains exactly one 1 and the rest 0, so all row sums and column sums equal 1. Each $P_{t,i}$ is a permutation matrix drawn uniformly from S_{2n} and is likewise doubly stochastic. The coefficients $\alpha^*, \alpha_{t,1}, \ldots, \alpha_{t,K_t}$ are strictly positive with $\alpha^* + \sum_i \alpha_{t,i} = 1$. Since $\mathcal{B}_{2n}$ is convex and D_t is a convex combination of elements of $\mathcal{B}_{2n}$:

$$D_t = \alpha^* M_t + \sum_{i=1}^{K_t} \alpha_{t,i} P_{t,i} \;\in\; \mathcal{B}_{2n}.$$

Applying the bilinear form $A \mapsto \mathbf{w}^T A\,\mathbf{y}$ to D_t and using linearity of matrix-vector multiplication:

$$\begin{aligned} f_t &= \mathbf{w}^T D_t\,\mathbf{y} \\ &= \mathbf{w}^T \left(\alpha^* M_t + \sum_{i=1}^{K_t} \alpha_{t,i} P_{t,i} \right) \mathbf{y} \\ &= \alpha^* \left(\mathbf{w}^T M_t\,\mathbf{y}\right) + \sum_{i=1}^{K_t} \alpha_{t,i} \left(\mathbf{w}^T P_{t,i}\,\mathbf{y}\right). \end{aligned} \tag{7}$$

By Lemma 3.9, $\mathbf{w}^T M_t \mathbf{y} = \sum_{j=1}^{n} b_{t,j} = s_t$. By definition (4), $\eta_t = \sum_{i=1}^{K_t} \alpha_{t,i} (\mathbf{w}^T P_{t,i} \mathbf{y})$. Substituting:

$$f_t = \alpha^*\, s_t + \eta_t.$$

Summing over all k clients:

$$\sum_{t=1}^{k} f_t = \sum_{t=1}^{k} (\alpha^*\, s_t + \eta_t) = \alpha^* \sum_{t=1}^{k} s_t + \sum_{t=1}^{k} \eta_t = \alpha^* S + \sum_{t=1}^{k} \eta_t.$$

The server knows $\sum_t f_t$ from the extraction step. The shuffled list $(\eta_{\pi(1)}, \ldots, \eta_{\pi(k)})$ is a permutation of $(\eta_1, \ldots, \eta_k)$, and the sum is invariant under permutation: $\sum_{j=1}^{k} \eta_{\pi(j)} = \sum_{t=1}^{k} \eta_t$. Therefore the server can compute:

$$S = \frac{1}{\alpha^*} \left(\sum_{t=1}^{k} f_t - \sum_{t=1}^{k} \eta_t \right) = \frac{1}{\alpha^*} \left(\alpha^* S + \sum_t \eta_t - \sum_t \eta_t \right) = \frac{1}{\alpha^*} \cdot \alpha^* S = S,$$

confirming that the protocol outputs the correct aggregate. □

4.5 Integrity in the Full Protocol

In the full (non-compressed) protocol, the server receives the matrix D_t directly and computes $f_t = \mathbf{w}^T D_t \mathbf{y}$ itself. A malicious client cannot cause the server to use an incorrect f_t because the server performs the extraction independently. Specifically, even if a client wished to inflate or deflate its contribution to the aggregate, the client can only control what matrix D_t it sends. The server then computes $f_t = \mathbf{w}^T D_t \mathbf{y}$ deterministically from D_t, so the client cannot make the server believe a different f_t than the one implied by the submitted D_t.

The server can additionally verify that the received D_t is a valid doubly stochastic matrix by checking that all entries are non-negative and that every row and column sums to 1 (within floating-point tolerance). If a client submits a matrix that is not doubly stochastic, the server can reject it. This verification does not reveal the client's private data (since any doubly stochastic matrix passes the check, regardless of which permutation is hidden inside), but it prevents malformed submissions that could corrupt the aggregate.

The remaining vulnerability is that a malicious client could submit a valid doubly stochastic matrix D_t that encodes a bitstream different from its true $\mathbf{b}_t$. This is the standard "input substitution" attack in the semi-honest model, where a dishonest client lies about its data. Preventing this requires mechanisms beyond the semi-honest model, such as zero-knowledge proofs that D_t is correctly constructed from the client's certified data source. We do not address this in the current work.

4.6 Compressed Variant

Algorithm 2 PolyVeil (Compressed)

Public parameter: $\alpha^* \in (0, 1)$, selection vectors $\mathbf{w}$, $\mathbf{y}$.
for each client $t \in \{1, \ldots, k\}$ in parallel **do**
  Compute $s_t = \sum_{j=1}^{n} b_{t,j}$ locally.
  Draw random permutations $P_{t,1}, \ldots, P_{t,K_t}$ and coefficients $\alpha_{t,i} > 0$ with $\sum_i \alpha_{t,i} = 1 - \alpha^*$.
  Compute $\eta_t = \sum_i \alpha_{t,i} \, (\mathbf{w}^T P_{t,i} \, \mathbf{y})$.
  Compute $f_t = \alpha^* s_t + \eta_t$.
  Send f_t to server.
end for
Clients jointly shuffle $\{\eta_1, \ldots, \eta_k\}$ and deliver to server.
Server computes $S = \frac{1}{\alpha^*}\big(\sum_t f_t - \sum_t \eta_t\big)$.

The compressed variant offers three concrete advantages beyond the communication reduction from $O(n^2)$ to $O(1)$ per client. First, the server never sees the doubly stochastic matrix D_t, which eliminates the BvN decomposition as an attack vector: the server has no matrix to decompose. Second, client-side computation drops from $O(n^2)$ (constructing a $2n \times 2n$ matrix) to $O(nK_t)$ (generating K_t random permutations and computing K_t bilinear forms). Third, server computation drops from $O(kn^2)$ to $O(k)$, becoming independent of the bitstream length.

The trade-off is that a malicious client can send an arbitrary f_t without the server being able to verify it, since the server no longer has D_t to check. Under the semi-honest model this is not a concern.

4.7 Worked Example with Full Computation

We trace every computation explicitly for $k = 3$ clients, $n = 2$ bits, $\alpha^* = 0.3$.

4.7.1 Ground Truth

Client 1 holds $\mathbf{b}_1 = (1, 0)$, so $s_1 = 1$. Client 2 holds $\mathbf{b}_2 = (1, 1)$, so $s_2 = 2$. Client 3 holds $\mathbf{b}_3 = (0, 1)$, so $s_3 = 1$. The true aggregate is $S = 1 + 2 + 1 = 4$.

The public parameters are $n = 2$, $\alpha^* = 0.3$, $\mathbf{w} = (1, 0, 1, 0)^T$, $\mathbf{y} = (0, 1, 0, 1)^T$.

4.7.2 Client-Side Masking (Full Detail for Client 1)

Encoding M_1. Client 1's bitstream is $(1, 0)$. Applying Definition 3.6:

$$M_1 = \text{blockdiag}(\Pi(1),\ \Pi(0)) = \text{blockdiag}\left(\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},\ \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\right) = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Generating decoy permutations. Client 1 chooses $K_1 = 2$ decoy permutations drawn uniformly from S_4 (the symmetric group on $\{1, 2, 3, 4\}$). Suppose the Fisher–Yates shuffle produces:

Permutation $\sigma_1 = (3, 1, 4, 2)$, meaning $\sigma_1(1) = 3$, $\sigma_1(2) = 1$, $\sigma_1(3) = 4$, $\sigma_1(4) = 2$. The corresponding permutation matrix has $P_{1,1}[i, \sigma_1(i)] = 1$:

$$P_{1,1} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix}.$$

Verification that this is correct — row 1 has its 1 in column 3 (since $\sigma_1(1) = 3$); row 2 in column 1; row 3 in column 4; row 4 in column 2. Each row and column has exactly one 1.

Permutation $\sigma_2 = (2, 1, 4, 3)$:

$$P_{1,2} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

To verify, row 1 has 1 in column 2; row 2 in column 1; row 3 in column 4; row 4 in column 3.

Choosing coefficients. Client 1 samples $\alpha_{1,1} = 0.5$ and $\alpha_{1,2} = 0.2$, satisfying $\alpha_{1,1} + \alpha_{1,2} = 0.7 = 1 - 0.3 = 1 - \alpha^*$.

Constructing D_1. Applying equation (3): $D_1 = \alpha^* M_1 + \alpha_{1,1} P_{1,1} + \alpha_{1,2} P_{1,2} = 0.3\, M_1 + 0.5\, P_{1,1} + 0.2\, P_{1,2}$.

Computing each scaled matrix:

$$0.3\, M_1 = \begin{pmatrix} 0 & 0.3 & 0 & 0 \\ 0.3 & 0 & 0 & 0 \\ 0 & 0 & 0.3 & 0 \\ 0 & 0 & 0 & 0.3 \end{pmatrix}, \quad 0.5\, P_{1,1} = \begin{pmatrix} 0 & 0 & 0.5 & 0 \\ 0.5 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.5 \\ 0 & 0.5 & 0 & 0 \end{pmatrix},$$

$$0.2\, P_{1,2} = \begin{pmatrix} 0 & 0.2 & 0 & 0 \\ 0.2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.2 \\ 0 & 0 & 0.2 & 0 \end{pmatrix}.$$

Summing entry by entry:

$$D_1 = \begin{pmatrix} 0+0+0 & 0.3+0+0.2 & 0+0.5+0 & 0+0+0 \\ 0.3+0.5+0.2 & 0+0+0 & 0+0+0 & 0+0+0 \\ 0+0+0 & 0+0+0 & 0.3+0+0 & 0+0.5+0.2 \\ 0+0+0 & 0+0.5+0 & 0+0+0.2 & 0.3+0+0 \end{pmatrix} = \begin{pmatrix} 0 & 0.5 & 0.5 & 0 \\ 1.0 & 0 & 0 & 0 \\ 0 & 0 & 0.3 & 0.7 \\ 0 & 0.5 & 0.2 & 0.3 \end{pmatrix}.$$

To verify that D_1 is doubly stochastic, the row sums are $0+0.5+0.5+0=1$, $1+0+0+0=1$, $0+0+0.3+0.7=1$, $0+0.5+0.2+0.3=1$. The column sums are $0+1+0+0=1$, $0.5+0+0+0.5=1$, $0.5+0+0.3+0.2=1$, $0+0+0.7+0.3=1$. All entries are non-negative.

Computing η_1. Applying equation (4): $\eta_1 = \alpha_{1,1}\,(\mathbf{w}^T P_{1,1}\,\mathbf{y}) + \alpha_{1,2}\,(\mathbf{w}^T P_{1,2}\,\mathbf{y})$.

For $P_{1,1}$:

$$P_{1,1}\mathbf{y} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 1 \end{pmatrix}, \qquad \mathbf{w}^T(P_{1,1}\mathbf{y}) = (1,0,1,0) \begin{pmatrix} 0 \\ 0 \\ 1 \\ 1 \end{pmatrix} = 0+0+1+0 = 1.$$

For $P_{1,2}$:

$$P_{1,2}\mathbf{y} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \qquad \mathbf{w}^T(P_{1,2}\mathbf{y}) = (1,0,1,0) \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix} = 1+0+1+0 = 2.$$

Therefore: $\eta_1 = 0.5 \times 1 + 0.2 \times 2 = 0.5 + 0.4 = 0.9$.

Client 1 sends D_1 to the server and holds $\eta_1 = 0.9$ for the shuffle.

4.7.3 Client-Side Masking for Clients 2 and 3 (Summary)

Client 2. Bitstream $\mathbf{b}_2 = (1,1)$, $s_2 = 2$.

$$M_2 = \text{blockdiag}(\Pi(1), \Pi(1)) = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

Suppose Client 2 draws $P_{2,1}$ corresponding to $\sigma = (4,3,2,1)$ and $P_{2,2}$ corresponding to $\sigma = (1,2,3,4)$ (the identity), with $\alpha_{2,1} = 0.4$, $\alpha_{2,2} = 0.3$:

$$P_{2,1} = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}, \qquad P_{2,2} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Then $D_2 = 0.3\,M_2 + 0.4\,P_{2,1} + 0.3\,P_{2,2}$. Compute $\mathbf{w}^T P_{2,1}\mathbf{y}$:

$$P_{2,1}\mathbf{y} = \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{w}^T \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix} = 2.$$

Compute $\mathbf{w}^T P_{2,2}\mathbf{y}$:

$$P_{2,2}\mathbf{y} = \mathbf{y} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix}, \quad \mathbf{w}^T\mathbf{y} = 0.$$

So $\eta_2 = 0.4 \times 2 + 0.3 \times 0 = 0.8$.

Client 3. Bitstream $\mathbf{b}_3 = (0, 1)$, $s_3 = 1$.

$$M_3 = \text{blockdiag}(\Pi(0), \Pi(1)) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

Suppose Client 3 draws $P_{3,1}$ from $\sigma = (2, 1, 3, 4)$ and $P_{3,2}$ from $\sigma = (3, 4, 1, 2)$, with $\alpha_{3,1} = 0.35$, $\alpha_{3,2} = 0.35$:

$$P_{3,1} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad P_{3,2} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}.$$

$P_{3,1}\mathbf{y} = (1, 0, 0, 1)^T$, so $\mathbf{w}^T(P_{3,1}\mathbf{y}) = 1 \cdot 1 + 0 \cdot 0 + 1 \cdot 0 + 0 \cdot 1 = 1$.

$P_{3,2}\mathbf{y} = (0, 1, 0, 1)^T$, so $\mathbf{w}^T(P_{3,2}\mathbf{y}) = 0$.

$\eta_3 = 0.35 \times 1 + 0.35 \times 0 = 0.35$.

4.7.4 Server-Side Extraction

The server computes $f_t = \mathbf{w}^T D_t\, \mathbf{y}$ for each client. By the decomposition proved in equation (7) of Theorem 4.1 (namely, that the bilinear extraction $\mathbf{w}^T D_t \mathbf{y}$ equals $\alpha^* s_t + \eta_t$ due to the linearity of the bilinear form and Lemma 3.9), we have:

For Client 1: $f_1 = \alpha^* s_1 + \eta_1 = 0.3 \times 1 + 0.9 = 1.2$.

We verify this by direct computation on D_1:

$$D_1\mathbf{y} = \begin{pmatrix} 0 & 0.5 & 0.5 & 0 \\ 1.0 & 0 & 0 & 0 \\ 0 & 0 & 0.3 & 0.7 \\ 0 & 0.5 & 0.2 & 0.3 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0.5 \\ 0 \\ 0.7 \\ 0.8 \end{pmatrix}.$$

$$f_1 = \mathbf{w}^T(D_1\mathbf{y}) = (1, 0, 1, 0)(0.5,\ 0,\ 0.7,\ 0.8)^T = 1 \cdot 0.5 + 0 \cdot 0 + 1 \cdot 0.7 + 0 \cdot 0.8 = 1.2.$$

For Client 2: $f_2 = \alpha^* s_2 + \eta_2 = 0.3 \times 2 + 0.8 = 1.4$.

For Client 3: $f_3 = \alpha^* s_3 + \eta_3 = 0.3 \times 1 + 0.35 = 0.65$.

4.7.5 Secure Noise Transmission

Each client submits its η_t to the shuffler: Client 1 submits $\eta_1 = 0.9$, Client 2 submits $\eta_2 = 0.8$, Client 3 submits $\eta_3 = 0.35$. The shuffler draws a uniformly random permutation $\pi \in S_3$; suppose $\pi = (2, 3, 1)$ (meaning $\pi(1) = 2$, $\pi(2) = 3$, $\pi(3) = 1$). The server receives the sequence $(\eta_{\pi(1)}, \eta_{\pi(2)}, \eta_{\pi(3)}) = (\eta_2, \eta_3, \eta_1) = (0.8,\ 0.35,\ 0.9)$. The server sees the values 0.8, 0.35, 0.9 but does *not* know that 0.8 came from Client 2, 0.35 from Client 3, and 0.9 from Client 1.

4.7.6 Aggregation

The server applies equation (6), which states $S = \frac{1}{\alpha^*}(\sum_{t=1}^{k} f_t - \sum_{t=1}^{k} \eta_t)$. This formula was derived in Theorem 4.1 from the fact that $f_t = \alpha^* s_t + \eta_t$ (equation (7)), so $\sum_t f_t - \sum_t \eta_t = \alpha^* \sum_t s_t$, and dividing by α^* recovers $\sum_t s_t$.

The server computes:

$$\sum_{t=1}^{3} f_t = 1.2 + 1.4 + 0.65 = 3.25.$$

$$\sum_{t=1}^{3} \eta_t = 0.8 + 0.35 + 0.9 = 2.05. \quad \text{(sum is the same regardless of shuffle order)}$$

$$S = \frac{1}{0.3}(3.25 - 2.05) = \frac{1.2}{0.3} = 4.$$

4.7.7 Verification

The ground truth is $S = s_1 + s_2 + s_3 = 1 + 2 + 1 = 4$. The protocol output matches. To see why the cancellation works algebraically:

$$\begin{aligned}
\sum_t f_t - \sum_t \eta_t &= (f_1 - \eta_1) + (f_2 - \eta_2) + (f_3 - \eta_3) \\
&\quad \text{(note: repairing the sum by the identity } f_t = \alpha^* s_t + \eta_t \text{ for each } t\text{)} \\
&= \alpha^* s_1 + \alpha^* s_2 + \alpha^* s_3 \\
&= \alpha^*(s_1 + s_2 + s_3) \\
&= 0.3 \times 4 = 1.2.
\end{aligned}$$

Dividing by $\alpha^* = 0.3$ gives 4. This cancellation holds for any values of the η_t's, any number of decoy permutations, and any positive coefficients, because it depends solely on the algebraic identity $f_t - \eta_t = \alpha^* s_t$.

4.7.8 What the Server Cannot Do — De-Shuffling Analysis

The server knows $f_1 = 1.2$, $f_2 = 1.4$, $f_3 = 0.65$ (linked to client identities) and the shuffled values $\{0.8, 0.35, 0.9\}$ (unlinked). To learn individual s_t values, the server must assign each shuffled η-value to the correct client. There are $3! = 6$ possible assignments (since the three values are distinct), and the server can test each:

Assignment σ	$\hat{s}_1 = \frac{f_1 - \eta_{\sigma(1)}}{\alpha^*}$	$\hat{s}_2 = \frac{f_2 - \eta_{\sigma(2)}}{\alpha^*}$	$\hat{s}_3 = \frac{f_3 - \eta_{\sigma(3)}}{\alpha^*}$	All $\hat{s}_t \in \{0, 1, 2\}$?
$(0.8, 0.35, 0.9)$	$\frac{0.4}{0.3} = 1.33$	$\frac{1.05}{0.3} = 3.50$	$\frac{-0.25}{0.3} = -0.83$	No
$(0.8, 0.9, 0.35)$	1.33	1.67	1.00	No
$(0.35, 0.8, 0.9)$	2.83	2.00	-0.83	No
$(0.35, 0.9, 0.8)$	2.83	1.67	-0.50	No
$(0.9, 0.35, 0.8)$	1.00	3.50	-0.50	No
$(0.9, 0.8, 0.35)$	**1.00**	**2.00**	**1.00**	**Yes**

In this example, only one assignment yields valid bit counts (integers in $\{0, \ldots, n\}$). This is a consequence of the small parameters ($n = 2$, $k = 3$). With larger n and k, multiple assignments will produce valid integer counts summing to S, and the server cannot distinguish among them. We analyze this formally in Section 5.

The probability that the server guesses the correct assignment is $1/k! = 1/6$ by random guessing alone. However, as we show in Section 5, the server can exploit the integrality constraint $s_t \in \{0, \ldots, n\}$ to identify the correct assignment with probability 1, rendering the naive protocol insecure.

To **prevent de-shuffling**, the protocol relies on the trusted shuffler $\mathcal{F}_{\text{shuffle}}$ (Section 4.2), which guarantees that the permutation π is uniformly random and unknown to the server. As we analyze rigorously in Section 5, this shuffling alone is *not sufficient* to prevent the server from recovering individual s_t values, because of an integrality constraint that enables deterministic de-shuffling. This motivates the protocol modifications presented in Section 8.

5 Security Analysis

We provide a rigorous security analysis of the basic protocol variants (Algorithms 1 and 2). We identify fundamental vulnerabilities in both variants, quantify the information leakage precisely, and defer the corrected protocols to Section 8.

5.1 The De-Shuffling Attack (Compressed Protocol)

The compressed protocol has the server receive identity-linked scalars $f_t = \alpha^* s_t + \eta_t$ for $t \in [k]$ and the shuffled sequence $(\tilde{\eta}_1, \ldots, \tilde{\eta}_k) = (\eta_{\pi(1)}, \ldots, \eta_{\pi(k)})$.

Theorem 5.1 (De-shuffling via integrality constraint)**.** *In the compressed protocol, the server can recover every client's bit count s_t with probability 1 (over the protocol's randomness).*

Proof. The server observes:

- Identity-linked scalars: $(1, f_1), (2, f_2), \ldots, (k, f_k)$ where $f_t = \alpha^* s_t + \eta_t$.
- Shuffled noise: $(\tilde{\eta}_1, \ldots, \tilde{\eta}_k) = (\eta_{\pi(1)}, \ldots, \eta_{\pi(k)})$ for unknown $\pi \in S_k$.

The server's test. For each candidate bijection $\sigma\colon [k] \to [k]$, the server computes:

$$\hat{s}_t(\sigma) \;=\; \frac{f_t - \tilde{\eta}_{\sigma(t)}}{\alpha^*}, \qquad t = 1, \ldots, k,$$

and checks whether $\hat{s}_t(\sigma) \in \{0, 1, \ldots, n\}$ for all t simultaneously.

The true assignment passes the test. Let $\sigma^* = \pi^{-1}$ be the true assignment (i.e., $\tilde{\eta}_{\sigma^*(t)} = \eta_t$ for all t). Then:

$$\hat{s}_t(\sigma^*) = \frac{f_t - \tilde{\eta}_{\sigma^*(t)}}{\alpha^*} = \frac{f_t - \eta_t}{\alpha^*} = \frac{(\alpha^* s_t + \eta_t) - \eta_t}{\alpha^*} = s_t.$$

Since $s_t = \sum_{j=1}^{n} b_{t,j} \in \{0, 1, \ldots, n\}$, all k candidate values pass the integrality test.

Any wrong assignment fails with probability 1. Let $\sigma \neq \sigma^*$. There exists some $t_0 \in [k]$ with $\sigma(t_0) \neq \sigma^*(t_0)$. Let $j = \pi(\sigma(t_0))$ be the client whose η-value is at position $\sigma(t_0)$ in the shuffled sequence, so $\tilde{\eta}_{\sigma(t_0)} = \eta_j$ with $j \neq t_0$. Then:

$$\hat{s}_{t_0}(\sigma) = \frac{f_{t_0} - \eta_j}{\alpha^*} = \frac{\alpha^* s_{t_0} + \eta_{t_0} - \eta_j}{\alpha^*} = s_{t_0} + \frac{\eta_{t_0} - \eta_j}{\alpha^*}. \tag{8}$$

This equals an integer if and only if $(\eta_{t_0} - \eta_j)/\alpha^*$ is an integer, i.e., $\eta_{t_0} - \eta_j \in \alpha^* \mathbb{Z}$.

Now we show $\Pr[\eta_{t_0} - \eta_j \in \alpha^* \mathbb{Z}] = 0$. Each $\eta_t = \sum_{i=1}^{K_t} \alpha_{t,i} X_{t,i}$ where $X_{t,i} = \mathbf{w}^T P_{t,i} \mathbf{y} \in \{0, 1, \ldots, n\}$ are integer-valued and the coefficients $\alpha_{t,i}$ are drawn from a continuous distribution on $\{(\alpha_1, \ldots, \alpha_{K_t}) : \alpha_i > 0, \sum_i \alpha_i = 1 - \alpha^*\}$. Consider the conditional distribution of η_{t_0} given all other randomness (including η_j and the integer values $X_{t_0,i}$). Conditional on $X_{t_0,1}, \ldots, X_{t_0,K_{t_0}}$, the

random variable $\eta_{t_0} = \sum_i \alpha_{t_0,i} X_{t_0,i}$ is a linear function of the continuously distributed coefficients $(\alpha_{t_0,1}, \ldots, \alpha_{t_0,K_{t_0}})$. Since the $X_{t_0,i}$ are not all equal (with probability 1, as the permutations are drawn independently and uniformly), this linear function is non-constant, so η_{t_0} has a continuous conditional distribution. Therefore:

$$\Pr[\eta_{t_0} - \eta_j \in \alpha^* \mathbb{Z}] = \Pr[\eta_{t_0} \in \eta_j + \alpha^* \mathbb{Z}] = 0,$$

since a continuous random variable assigns probability zero to any countable set.

Uniqueness and conclusion. Since σ^* passes the test and each $\sigma \neq \sigma^*$ fails with probability 1, taking a union bound over the (finite) set of $k! - 1$ wrong assignments:

$$\Pr[\text{any wrong } \sigma \text{ passes}] \leq \sum_{\sigma \neq \sigma^*} \Pr[\sigma \text{ passes}] = 0.$$

Therefore, with probability 1, the server uniquely identifies σ^* and recovers $s_t = \hat{s}_t(\sigma^*)$ for all $t \in [k]$. □

Remark 5.2 (This attack is demonstrated in the worked example)**.** *The worked example in Section 4.7 illustrates this attack explicitly. The server tests all* $3! = 6$ *assignments of shuffled* η *values to clients and finds that exactly one assignment yields valid bit counts* $(s_1, s_2, s_3) = (1, 2, 1)$ *with all* $s_t \in \{0, 1, 2\}$*. The paper previously described this as a special case for small* n*, but Theorem 5.1 shows it works for all* n*.*

5.2 Direct Inference from the Marginal (Compressed Protocol)

Even without de-shuffling, the server can perform Bayesian inference on s_t from f_t alone.

Proposition 5.3 (Server's posterior from f_t)**.** *Let* μ *denote the probability density of* η_t *(which is continuous and independent of* $\mathbf{b}_t$*). The server's posterior distribution over* s_t *given* f_t *is:*

$$\Pr(s_t = s \mid f_t) = \frac{\mu(f_t - \alpha^* s) \cdot \pi(s)}{\sum_{s'=0}^{n} \mu(f_t - \alpha^* s') \cdot \pi(s')}, \tag{9}$$

where $\pi(s)$ *is any prior distribution over* $s_t \in \{0, 1, \ldots, n\}$*.*

Proof. We derive this from Bayes' theorem. The server knows f_t and seeks s_t. Since $f_t = \alpha^* s_t + \eta_t$ and η_t is independent of s_t with density μ, the conditional density of f_t given $s_t = s$ is:

$$p(f_t \mid s_t = s) = p(\alpha^* s + \eta_t = f_t) = p(\eta_t = f_t - \alpha^* s) = \mu(f_t - \alpha^* s).$$

The second equality uses the deterministic relationship $f_t = \alpha^* s + \eta_t$, so fixing $s_t = s$ means $\eta_t = f_t - \alpha^* s$. Applying Bayes' theorem with prior $\pi(s) = \Pr(s_t = s)$:

$$\Pr(s_t = s \mid f_t) = \frac{p(f_t \mid s_t = s) \cdot \pi(s)}{p(f_t)} = \frac{\mu(f_t - \alpha^* s) \cdot \pi(s)}{\sum_{s'=0}^{n} \mu(f_t - \alpha^* s') \cdot \pi(s')},$$

where the denominator is the marginal density $p(f_t) = \sum_{s'=0}^{n} p(f_t \mid s_t = s')\pi(s')$, obtained by the law of total probability over the $(n+1)$ possible values of s_t. □

The MAP (maximum a posteriori) estimator selects the s that maximizes the numerator. For a uniform prior, this reduces to $\hat{s}_t = \arg\max_s \mu(f_t - \alpha^* s)$.

Remark 5.4 (This channel is weak for small α^*, but irrelevant given de-shuffling)**.** *When α^* is small relative to the standard deviation of η_t, the density $\mu(f_t - \alpha^* s)$ varies slowly with s and the posterior is nearly uniform. To see this, we compute* $\mathrm{std}(\eta_t)$.

Each term $\mathbf{w}^T P_{t,i}\mathbf{y} = \sum_{j=1}^{n} (P_{t,i})_{2j-1,2j}$, which counts the number of 2×2 diagonal blocks in which $P_{t,i}$ maps the odd index to the even index. For a uniform random permutation matrix, each $(P_{t,i})_{2j-1,2j}$ is a Bernoulli random variable, and $\mathbf{w}^T P_{t,i}\mathbf{y} \in \{0, 1, \ldots, n\}$. Its expectation is:

$$\mathbb{E}[\mathbf{w}^T P_{t,i}\mathbf{y}] = \sum_{j=1}^{n} \mathbb{E}[(P_{t,i})_{2j-1,2j}] = \sum_{j=1}^{n} \frac{1}{2n} = \frac{n}{2n} = \frac{1}{2}.$$

To see this, note that $\mathbf{w}^T P\mathbf{y}$ extracts the sum of entries at positions $(1,2), (3,4), (5,6), \ldots, (2n-1, 2n)$. For a uniform random permutation σ, $(P)_{2j-1,2j} = \mathbf{1}[\sigma(2j-1) = 2j]$, and $\Pr[\sigma(2j-1) = 2j] = 1/(2n)$. Summing over $j = 1, \ldots, n$ gives $n/(2n) = 1/2$. The variance involves correlations between blocks, but for large n is approximately $n \cdot \frac{1}{2n}(1 - \frac{1}{2n}) \approx 1/2$. Therefore $\mathrm{Var}[\mathbf{w}^T P\mathbf{y}] = \Theta(1)$ and $\mathrm{std}[\mathbf{w}^T P\mathbf{y}] = \Theta(1)$.

Now $\eta_t = \sum_{i=1}^{K_t} \alpha_{t,i}(\mathbf{w}^T P_{t,i}\mathbf{y})$. Its mean is:

$$\mathbb{E}[\eta_t] = \sum_{i=1}^{K_t} \alpha_{t,i} \cdot \frac{1}{2} = \frac{1 - \alpha^*}{2}.$$

Its variance (conditional on coefficients, using independence of the $P_{t,i}$) is:

$$\mathrm{Var}[\eta_t] = \sum_{i=1}^{K_t} \alpha_{t,i}^2 \cdot \mathrm{Var}[\mathbf{w}^T P_{t,i}\mathbf{y}] = \Theta(1) \cdot \sum_{i=1}^{K_t} \alpha_{t,i}^2.$$

For K_t roughly uniform weights, $\sum_i \alpha_{t,i}^2 \approx (1-\alpha^*)^2/K_t$, giving $\mathrm{std}[\eta_t] = \Theta((1-\alpha^*)/\sqrt{K_t})$.

The signal (difference between f_t for adjacent s values) is α^*. The total signal range across all $n+1$ possible s_t values is $\alpha^* n$. The signal-to-noise ratio for distinguishing s_t from $s_t + 1$ is:

$$\mathrm{SNR} = \frac{\alpha^*}{\mathrm{std}[\eta_t]} = \frac{\alpha^* \sqrt{K_t}}{(1-\alpha^*) \cdot \Theta(1)}.$$

For $\alpha^* = 1/(4n)$ and $K_t = O(1)$, this is $O(1/n)$, making the per-unit SNR negligible.

However, this analysis is moot because the de-shuffling attack of Theorem 5.1 recovers s_t with probability 1 regardless of the signal-to-noise ratio.

5.3 Likelihood Analysis of the Full Protocol

In the full protocol, the server additionally receives $D_t = \alpha^* M_t + (1-\alpha^*)R_t$, where $R_t \sim \nu$ is the random decoy component. The server can compute the likelihood of each candidate permutation matrix M':

Proposition 5.5 (Posterior concentration in the full protocol)**.** *Let ν denote the distribution of $R_t = \frac{1}{1-\alpha^*} \sum_i \alpha_{t,i} P_{t,i}$ over $\mathcal{B}_{2n}$. For each candidate $M' \in \mathcal{L}(D_t, \alpha^*)$, the likelihood is:*

$$\Pr(D_t \mid M_t = M') = \frac{1}{(1-\alpha^*)^d}\, \nu\!\left(\frac{D_t - \alpha^* M'}{1-\alpha^*}\right),$$

where d is the dimension of $\mathcal{B}_{2n}$. As $K_t \to \infty$, the posterior probability of the true M_t approaches 1.

Proof. Deriving the likelihood. The masked matrix is $D_t = \alpha^* M_t + (1-\alpha^*)R_t$, where $R_t \sim \nu$. For a candidate M', define the residual $R' = (D_t - \alpha^* M')/(1-\alpha^*)$. We want $\Pr(D_t \mid M_t = M')$. Since $D_t = \alpha^* M' + (1-\alpha^*)R_t$ when $M_t = M'$, the event $\{D_t = d\}$ is the event $\{R_t = (d - \alpha^* M')/(1-\alpha^*)\}$. The map $\varphi\colon R \mapsto \alpha^* M' + (1-\alpha^*)R$ is an affine transformation from $\mathcal{B}_{2n}$ to itself. Its Jacobian matrix is $(1-\alpha^*)I_d$ where $d = (2n-1)^2$ is the dimension of $\mathcal{B}_{2n}$ (a doubly stochastic matrix has $(2n)^2$ entries but $(2n-1)^2$ degrees of freedom after enforcing row and column sum constraints). The absolute value of the Jacobian determinant is $|\det((1-\alpha^*)I_d)| = (1-\alpha^*)^d$. By the standard change-of-variables formula for densities, if R_t has density $\nu(R)$, then $D_t = \varphi(R_t)$ has density:

$$p(d \mid M_t = M') = \frac{1}{(1-\alpha^*)^d}\,\nu\!\left(\frac{d - \alpha^* M'}{1-\alpha^*}\right) = \frac{1}{(1-\alpha^*)^d}\,\nu(R').$$

The true residual was drawn from ν; false residuals were not. For the true M_t, the residual is:

$$R = \frac{D_t - \alpha^* M_t}{1-\alpha^*} = \frac{(\alpha^* M_t + (1-\alpha^*)R_t) - \alpha^* M_t}{1-\alpha^*} = R_t,$$

which was drawn from ν by construction. For any $M' \neq M_t$:

$$\begin{aligned} R' &= \frac{D_t - \alpha^* M'}{1-\alpha^*} = \frac{(\alpha^* M_t + (1-\alpha^*)R_t) - \alpha^* M'}{1-\alpha^*} \\ &= R_t + \frac{\alpha^*(M_t - M')}{1-\alpha^*} \\ &= R + \frac{\alpha^*}{1-\alpha^*}(M_t - M'). \end{aligned}$$

Since $M_t \neq M'$, the matrix $M_t - M'$ is nonzero (it has entries in $\{-1, 0, 1\}$ with at least two nonzero entries), so R' is a translate of R by a fixed nonzero shift.

Computing $\mathbb{E}[P_{ij}]$ for a uniform random permutation matrix. Let P be a uniform random permutation matrix in S_{2n}, corresponding to a permutation σ drawn uniformly from the symmetric group S_{2n}. The (i,j) entry of P is the indicator $P_{ij} = \mathbf{1}[\sigma(i) = j]$. Since σ is uniform over all $(2n)!$ permutations:

$$\mathbb{E}[P_{ij}] = \Pr[\sigma(i) = j] = \frac{|\{\sigma \in S_{2n} : \sigma(i) = j\}|}{|S_{2n}|} = \frac{(2n-1)!}{(2n)!} = \frac{1}{2n}.$$

The numerator counts the permutations that map i to j: once $\sigma(i) = j$ is fixed, the remaining $2n-1$ elements can be mapped in $(2n-1)!$ ways. Therefore:

$$\mathbb{E}[P] = \frac{1}{2n}\mathbf{J},$$

where $\mathbf{J}$ is the $(2n) \times (2n)$ all-ones matrix. (One can verify: each row of $\mathbb{E}[P]$ sums to $(2n) \cdot \frac{1}{2n} = 1$, consistent with P being doubly stochastic.)

Computing $\mathrm{Var}[P_{ij}]$. Since $P_{ij} = \mathbf{1}[\sigma(i) = j] \in \{0, 1\}$ is a Bernoulli random variable with parameter $p = 1/(2n)$:

$$\mathrm{Var}[P_{ij}] = p(1-p) = \frac{1}{2n}\left(1 - \frac{1}{2n}\right) = \frac{2n-1}{(2n)^2}.$$

Concentration of R_t as $K_t \to \infty$. Each entry of R_t is:

$$(R_t)_{ab} = \frac{1}{1-\alpha^*}\sum_{i=1}^{K_t} \alpha_{t,i}(P_{t,i})_{ab}.$$

Its expectation is:

$$\mathbb{E}[(R_t)_{ab}] = \frac{1}{1-\alpha^*}\sum_{i=1}^{K_t}\alpha_{t,i}\,\mathbb{E}[(P_{t,i})_{ab}] = \frac{1}{1-\alpha^*}\sum_{i=1}^{K_t}\alpha_{t,i}\cdot\frac{1}{2n} = \frac{1}{1-\alpha^*}\cdot\frac{1}{2n}\sum_{i=1}^{K_t}\alpha_{t,i} = \frac{1}{1-\alpha^*}\cdot\frac{1}{2n}\cdot(1-\alpha^*) = \frac{1}{2n},$$

where we used $\sum_{i=1}^{K_t}\alpha_{t,i} = 1-\alpha^*$.

Its variance is (using independence of the $P_{t,i}$ and treating $\alpha_{t,i}$ as fixed conditional on the coefficient draw):

$$\mathrm{Var}[(R_t)_{ab}] = \frac{1}{(1-\alpha^*)^2}\sum_{i=1}^{K_t}\alpha_{t,i}^2\,\mathrm{Var}[(P_{t,i})_{ab}] = \frac{1}{(1-\alpha^*)^2}\sum_{i=1}^{K_t}\alpha_{t,i}^2\cdot\frac{2n-1}{(2n)^2}.$$

For roughly uniform weights $\alpha_{t,i} \approx (1-\alpha^*)/K_t$, we have $\sum_i \alpha_{t,i}^2 \approx K_t\cdot((1-\alpha^*)/K_t)^2 = (1-\alpha^*)^2/K_t$. Substituting:

$$\mathrm{Var}[(R_t)_{ab}] \approx \frac{1}{(1-\alpha^*)^2}\cdot\frac{(1-\alpha^*)^2}{K_t}\cdot\frac{2n-1}{(2n)^2} = \frac{2n-1}{(2n)^2K_t} = \frac{1}{K_t}\cdot\frac{1}{2n}\left(1-\frac{1}{2n}\right).$$

The standard deviation in each entry is therefore $\mathrm{std}[(R_t)_{ab}] = O(1/\sqrt{K_t})$. By Chebyshev's inequality applied entry-wise, for any $\varepsilon > 0$:

$$\Pr\left[\left|(R_t)_{ab} - \frac{1}{2n}\right| > \varepsilon\right] \le \frac{\mathrm{Var}[(R_t)_{ab}]}{\varepsilon^2} = O\left(\frac{1}{K_t\varepsilon^2}\right) \to 0 \quad \text{as } K_t \to \infty.$$

Hence $R_t \to \frac{1}{2n}\mathbf{J}$ in probability, entry-wise.

Likelihood ratio diverges. The true residual R satisfies $\|R - \frac{1}{2n}\mathbf{J}\|_\infty = O_P(1/\sqrt{K_t})$ (where $\|\cdot\|_\infty$ is the max entry). The false residual has:

$$\left\|R' - \frac{1}{2n}\mathbf{J}\right\|_\infty = \left\|R - \frac{1}{2n}\mathbf{J} + \frac{\alpha^*}{1-\alpha^*}(M_t - M')\right\|_\infty.$$

The shift matrix $\frac{\alpha^*}{1-\alpha^*}(M_t - M')$ has entries of magnitude $\frac{\alpha^*}{1-\alpha^*}$ at the positions where M_t and M' differ. Since $M_t \neq M'$ (they differ in at least 2 rows), we have:

$$\left\|R' - \frac{1}{2n}\mathbf{J}\right\|_\infty \ge \frac{\alpha^*}{1-\alpha^*} - O_P\left(\frac{1}{\sqrt{K_t}}\right).$$

For K_t large enough that $1/\sqrt{K_t} \ll \alpha^*/(1-\alpha^*)$, the false residual R' lies at distance $\Theta(\alpha^*/(1-\alpha^*))$ from the mode $\frac{1}{2n}\mathbf{J}$, while R lies at distance $O(1/\sqrt{K_t})$. Since ν concentrates with width $O(1/\sqrt{K_t})$, the false residual is $\Theta(\alpha^*\sqrt{K_t}/(1-\alpha^*))$ standard deviations from the mode. Therefore $\nu(R')/\nu(R) \to 0$ as $K_t \to \infty$ (assuming α^* is fixed), and the likelihood ratio diverges:

$$\frac{\nu(R)}{\nu(R')} \;\to\; \infty \qquad \text{as } K_t \to \infty.$$

Applying Bayes' theorem with a uniform prior $\pi(M') = 1/(2n)!$ for all $M' \in S_{2n}$:

$$\Pr(M_t = M' \mid D_t) = \frac{\nu(R')}{\sum_{M''\in S_{2n}}\nu(R'')} \;\to\; \begin{cases} 1 & \text{if } M' = M_t, \\ 0 & \text{if } M' \neq M_t, \end{cases}$$

since the numerator for $M' = M_t$ dominates all other terms. □

Remark 5.6 (More decoys can decrease security)**.** *This creates a counterintuitive trade-off. Increasing K_t (adding more decoy permutations) was intended to improve security by increasing the decomposition count L. However, increasing K_t also concentrates ν, making the likelihood ratio $\nu(R)/\nu(R')$ larger and the server's MAP estimate more accurate. Against a computationally unbounded adversary, the concentration effect dominates the decomposition-count effect.*

5.4 Summary of Vulnerabilities in the Basic Protocol

Attack	**Compressed**	**Full**
De-shuffling (integrality)	Recovers all s_t w.p. 1	Recovers all s_t w.p. 1
Marginal inference (f_t alone)	Weak for small α^*	Weak for small α^*
Likelihood ratio (D_t alone)	N/A	Strong for large K_t

The de-shuffling attack is decisive: both the compressed and full protocols, as described in Algorithms 1 and 2, reveal all individual s_t values to an honest-but-curious server with probability 1. The root cause is that the server receives identity-linked f_t values *and* the shuffled η_t values *separately*, and the integrality of s_t provides a verification criterion that uniquely identifies the true shuffle permutation.

This vulnerability motivates the protocol modifications in Section 8, which eliminate the de-shuffling channel by altering what the server receives.

6 Geometry of the Protocol Parameter α^*

The Birkhoff polytope $\mathcal{B}_{2n}$ is a convex body whose vertices are the $(2n)!$ permutation matrices. Its *boundary* consists of doubly stochastic matrices with at least one zero entry. Its *interior* $\mathrm{int}(\mathcal{B}_{2n})$ consists of matrices with all entries strictly positive. A matrix "deep in the interior" has its minimum entry bounded away from zero, meaning it is geometrically far from every face.

The masked matrix $D_t = \alpha^* M_t + (1-\alpha^*)R_t$ lies on the line segment in $\mathcal{B}_{2n}$ from the random interior point $R_t = \frac{1}{1-\alpha^*}\sum_i \alpha_{t,i} P_{t,i}$ to the vertex M_t, at parameter α^* of the way from R_t toward M_t.

Large α^* (close to 1). $D_t \approx M_t$ lies near a vertex. Most entries are close to zero, revealing the structure of M_t. The number of consistent alternative permutations $L(D_t, \alpha^*)$ is small because $(D_t)_{ij} < \alpha^*$ at many positions.

Small α^* (close to 0). $D_t \approx R_t$ lies in the interior with all entries positive. The condition $(D_t)_{ij} \geq \alpha^*$ holds everywhere, giving $L(D_t, \alpha^*) = (2n)!$.

The precise condition. For every permutation matrix M' to be consistent with D_t, we need $(D_t)_{ij} \geq \alpha^*$ wherever $M'_{ij} = 1$. Since every position (i,j) has $M'_{ij} = 1$ for some permutation, the condition is $\alpha^* \leq \min_{ij}(D_t)_{ij}$. Since $\min_{ij}(D_t)_{ij} \geq (1-\alpha^*)\min_{ij}(R_t)_{ij}$, a sufficient condition is $\alpha^* \leq \frac{\min_{ij}(R_t)_{ij}}{1+\min_{ij}(R_t)_{ij}}$.

Proposition 6.1. *The condition $\alpha^* \leq \min_{ij}(D_t)_{ij}$ holds with probability at least $1-\delta$ when $K_t \geq 2n\ln(4n^2/\delta)$ and $\alpha^* \leq \min_l \alpha_{t,l}$.*

Proof. We proceed in three steps: bound the probability of zero entries in R_t, derive the minimum entry of D_t, and combine.

Probability that $(R_t)_{ij} = 0$. The decoy component is $R_t = \frac{1}{1-\alpha^*}\sum_{l=1}^{K_t} \alpha_{t,l} P_{t,l}$. Entry (i,j) satisfies $(R_t)_{ij} = 0$ if and only if $(P_{t,l})_{ij} = 0$ for every $l \in [K_t]$ (since $\alpha_{t,l} > 0$). Each $P_{t,l}$ is a uniform

random permutation matrix in S_{2n}, so:

$$\Pr[(P_{t,l})_{ij} = 0] = 1 - \frac{1}{2n},$$

since $P_{t,l}$ places exactly one 1 per row, and column j is chosen uniformly from $2n$ columns. The decoys are independent across l, so:

$$\Pr[(R_t)_{ij} = 0] = \prod_{l=1}^{K_t} \Pr[(P_{t,l})_{ij} = 0] = \left(1 - \frac{1}{2n}\right)^{K_t}.$$

Union bound over all entries. We want $\min_{ij}(R_t)_{ij} > 0$, i.e., no entry of R_t is zero. The complementary event is $\{\exists\,(i,j) : (R_t)_{ij} = 0\}$. By a union bound over all $(2n)^2$ entries:

$$\Pr\left[\min_{ij}(R_t)_{ij} = 0\right] \le \sum_{i=1}^{2n}\sum_{j=1}^{2n} \Pr[(R_t)_{ij} = 0] = (2n)^2 \left(1 - \frac{1}{2n}\right)^{K_t}.$$

Using the standard inequality $1 - x \le e^{-x}$ for $x = 1/(2n) > 0$:

$$\left(1 - \frac{1}{2n}\right)^{K_t} \le e^{-K_t/(2n)}.$$

Therefore:

$$\Pr\left[\min_{ij}(R_t)_{ij} = 0\right] \le 4n^2 \exp\left(-\frac{K_t}{2n}\right).$$

Setting this to at most δ and solving: $4n^2 e^{-K_t/(2n)} \le \delta \iff K_t \ge 2n \ln(4n^2/\delta)$.

From R_t positive to $\alpha^* \le \min_{ij}(D_t)_{ij}$. Conditional on all entries of R_t being positive, the minimum entry of D_t is:

$$\min_{ij}(D_t)_{ij} = \min_{ij}\left(\alpha^*(M_t)_{ij} + \sum_{l=1}^{K_t} \alpha_{t,l}(P_{t,l})_{ij}\right).$$

Since $(M_t)_{ij} \ge 0$, we have $(D_t)_{ij} \ge \sum_l \alpha_{t,l}(P_{t,l})_{ij}$. When $(R_t)_{ij} > 0$, there exists at least one l_0 with $(P_{t,l_0})_{ij} = 1$, contributing α_{t,l_0} to the sum. Therefore:

$$\min_{ij}(D_t)_{ij} \ge \min_{ij} \sum_l \alpha_{t,l}(P_{t,l})_{ij} \ge \min_l \alpha_{t,l}.$$

Hence $\alpha^* \le \min_l \alpha_{t,l}$ implies $\alpha^* \le \min_{ij}(D_t)_{ij}$, which is the required interior condition. □

Practical guidance. For $n = 1000$, setting $K_t \approx 30{,}000$ and $\alpha^* = 1/(4n) = 0.00025$ ensures the condition with probability exceeding $1 - 10^{-6}$. Very small α^* (e.g., 10^{-15}) causes floating-point errors in the subtraction $\sum f_t - \sum \eta_t$. A practical range is $\alpha^* \in [1/(4n), 1/(2n)]$.

7 Analysis of Alternatives, Communication, and Attack Resilience

7.1 Positioning Combinatorial Privacy

PolyVeil instantiates a paradigm we call Combinatorial Privacy, which complements MPC, HE, and DP. In this paradigm, private data is embedded in a combinatorial structure (the Birkhoff polytope) whose decomposition is provably non-unique. The protocol uses a single lightweight cryptographic primitive (a shuffler) for aggregation, but the core privacy mechanism is combinatorial. PolyVeil cannot compute arbitrary functions (unlike MPC), does not operate on ciphertexts (unlike HE), and produces exact answers (unlike DP).

7.2 Comparison with Generic Secure MPC

Circuit complexity of the Boolean sum. To understand the concrete cost of MPC for this problem, we must first determine the circuit that computes the Boolean sum $S = \sum_{t=1}^{k} s_t = \sum_{t=1}^{k} \sum_{j=1}^{n} b_{t,j}$. This is a sum of $N = kn$ bits, producing an output in $\{0, 1, \ldots, N\}$ representable in $\lceil \log_2(N+1) \rceil$ bits. An addition tree with $N-1$ adder stages, each adding two $\lceil \log_2 N \rceil$-bit numbers, requires $O(N \log N) = O(kn \log(kn))$ Boolean gates. This is the circuit size $|C|$ for this problem.

In garbled-circuit-based MPC (e.g., Yao's protocol for two parties, or its multi-party extensions), each Boolean gate requires one or more oblivious transfers (OTs), each costing $O(\kappa)$ bits of communication where κ is the security parameter (typically 128). The total communication is therefore $O(|C| \cdot \kappa) = O(kn \log(kn) \cdot \kappa)$ per pair of parties, and in a k-party setting the naive approach requires $O(k^2)$ pairwise executions, giving total communication $O(k^2 n \log(kn) \cdot \kappa)$. Even with OT extension techniques that amortize the per-OT cost, the circuit-proportional overhead remains.

In secret-sharing-based MPC (e.g., SPDZ or its variants), each addition gate is "free" (requiring only local computation) but each multiplication gate requires a preprocessed Beaver triple and $O(k\kappa)$ bits of communication. For the Boolean sum, the circuit consists entirely of additions if the inputs are in an arithmetic field, making the online phase very cheap at $O(k)$ communication. However, the offline phase for generating Beaver triples (needed for the multiplication gates in the binary-to-arithmetic conversion) and the input-sharing phase still require $O(kn\kappa)$ communication. Furthermore, secret-sharing MPC requires an interactive setup round where each client secret-shares its input among all parties, costing $O(k^2 n)$ communication.

PolyVeil's advantage. In PolyVeil's compressed variant, each client sends a single scalar f_t to the server and a single scalar η_t to the shuffler. The total communication is $O(k)$ real numbers, independent of n, κ, and any circuit complexity. For a concrete comparison, consider $k = 1000$ clients with $n = 10{,}000$ bits each at security parameter $\kappa = 128$. The garbled-circuit MPC approach would require on the order of $k^2 n \log(kn)\kappa \approx 10^6 \times 10^4 \times 24 \times 128 \approx 3 \times 10^{13}$ bits ≈ 3.8 terabytes of total communication. Secret-sharing approaches reduce this substantially but still require $O(kn\kappa) \approx 10^3 \times 10^4 \times 128 \approx 1.3 \times 10^9$ bits ≈ 160 megabytes for input sharing alone. PolyVeil's compressed variant requires $2k = 2000$ scalars ≈ 16 kilobytes. The gap is several orders of magnitude.

PolyVeil is not MPC. Despite the superficial similarity (multiple parties compute a joint function while preserving privacy), PolyVeil is not an MPC protocol in the standard sense. MPC protocols achieve security through cryptographic operations applied to the computation itself (garbling gates, sharing secrets, generating correlated randomness), and they can compute *any* function. PolyVeil achieves security through a different mechanism entirely — the combinatorial structure of the Birkhoff polytope — and is limited to summation (and linear functions more broadly). It does not garble circuits, share secrets among parties, or perform homomorphic operations. The only cryptographic component is the shuffler, which is a *communication primitive* rather than a computational one.

A more precise characterization is that PolyVeil is an instance of the Combinatorial Privacy paradigm, in which privacy derives from embedding private data in a combinatorial object with non-unique decompositions. It belongs to the broader family of protocols that use structured noise and anonymous communication to achieve privacy, alongside the shuffle model of differential privacy and mixnet-based anonymous messaging. The novelty of PolyVeil within this family is the use of the Birkhoff polytope's geometry to generate the structured noise.

When to prefer MPC over PolyVeil. MPC is the right choice when the target function is more complex than summation (e.g., computing a median, training a model, or evaluating a decision tree), when security against malicious adversaries is required, or when the protocol must compose securely with other cryptographic primitives in a larger system (UC security). For the specific problem of private Boolean sums in the semi-honest model, PolyVeil is more efficient by orders of magnitude.

7.3 Comparison with Differential Privacy

The fundamental difference is accuracy. In the local DP model, where each client randomizes their data independently, the error for estimating a Boolean sum over kn bits scales as $\Theta(\sqrt{kn}/\varepsilon)$, which grows with the number of clients. In the central model, where a trusted curator adds noise to the true aggregate, the error is $\Theta(n/\varepsilon)$, independent of k but still nonzero. The shuffle model interpolates, achieving central-model accuracy with local-model trust assumptions. PolyVeil produces exact results with zero error. When an application requires the precise count (e.g., regulatory reporting, election tallying, financial reconciliation), no DP mechanism can substitute for PolyVeil.

The trust assumptions of PolyVeil and shuffle-model DP are essentially identical, as both require a shuffler. The difference is where the privacy comes from. In shuffle-model DP, privacy arises from statistical noise combined with shuffling anonymity. In PolyVeil, privacy arises from algebraic masking (the BvN non-uniqueness) combined with shuffling anonymity. PolyVeil's approach is deterministic and exact, while shuffle-model DP is inherently randomized and approximate.

A further advantage of PolyVeil over central DP is that the server never observes individual client data, not even before noise addition. In central DP, the curator sees all raw data and must be trusted to add noise correctly. A compromised or dishonest curator can simply omit the noise and read the data in the clear. In PolyVeil, the server sees only masked doubly stochastic matrices (full protocol) or scalar f_t values (compressed protocol), and no individual bitstream is ever exposed to any party other than the client that holds it.

PolyVeil is forward-compatible with DP. After computing the exact aggregate S via PolyVeil, the server can add discrete Gaussian noise $Z \sim \mathcal{N}_{\mathbb{Z}}(0, \sigma^2)$ with $\sigma \geq n\sqrt{2\ln(1.25/\delta)}/\varepsilon$ to produce a privatized output $\tilde{S} = S + Z$ satisfying (ε, δ)-DP. The sensitivity of S with respect to replacing one client's bitstream is n (a client can change from all zeros to all ones), giving the standard Gaussian mechanism calibration. This composition provides two layers of protection: PolyVeil's cryptographic guarantee protects individual data during computation, and the DP guarantee protects the aggregate at release. The two layers compose cleanly because, by the simulation proof, the server's intermediate view is a deterministic function of S (in distribution), and adding noise to S is post-processing.

7.4 Comparison with Homomorphic Encryption

Paillier encryption provides exact results with IND-CPA security and requires no shuffler. Each client encrypts its bit count s_t under a common public key (one ciphertext of 2κ bits, e.g., 4096 bits at $\kappa = 128$), the server multiplies the k ciphertexts ($O(k)$ modular multiplications), and a designated decryption party recovers the sum. The total communication is $O(k\kappa)$ bits for the ciphertexts plus $O(\kappa)$ for the result.

PolyVeil's compressed variant requires $O(k)$ scalars of 64 bits each, giving $O(64k)$ bits total. For Paillier, the cost is $O(4096k)$ bits, a factor of 64 larger. This bandwidth gap may or may not be significant depending on the deployment, but it illustrates the lightweight nature of PolyVeil's

communication.

The more important distinction concerns infrastructure requirements. Paillier encryption requires a *public-key infrastructure* (PKI), meaning that someone must generate the key pair (pk, sk), distribute pk to all clients, and safeguard sk. If the decryption key sk is held by a single party and that party is compromised (by an attacker, a subpoena, or an insider), all past and future encrypted values can be decrypted, retrospectively destroying the privacy of every client. Threshold variants distribute sk among multiple parties to mitigate this risk, but they add significant protocol complexity (distributed key generation, threshold decryption rounds, handling of corrupted shareholders).

PolyVeil requires no PKI at all. There are no encryption keys, no decryption keys, no key distribution ceremonies, and no key storage. The only trust infrastructure is the shuffler, which processes ephemeral scalar values and need not maintain any long-term secrets. If the shuffler is compromised after the protocol completes, no retroactive damage occurs (the shuffler does not hold any key that could decrypt past sessions). This absence of long-term secrets is a significant operational advantage in deployment scenarios where key management is expensive, error-prone, or politically contentious (e.g., multi-jurisdictional health data aggregation where no single entity is trusted to hold a decryption key).

Combinatorial Privacy as a distinct paradigm. MPC and HE are both *cryptographic* paradigms that protect data by transforming it through computationally hard operations (oblivious transfer, modular exponentiation, lattice-based encryption). Their security relies on computational assumptions (factoring, discrete logarithm, learning with errors). DP is a *statistical* paradigm that protects data by adding calibrated random noise, with security defined information-theoretically through the (ε, δ) guarantee.

Combinatorial Privacy, the paradigm instantiated by PolyVeil, complements both the cryptographic and statistical approaches. It protects data by embedding it in a convex polytope (the Birkhoff polytope) where the non-uniqueness of the vertex decomposition means that any observed doubly stochastic matrix is consistent with multiple alternative private inputs. In the corrected two-layer protocol (Section 8), the main server achieves information-theoretic security via aggregation, while the aggregation entity faces a computational barrier from the #P-hardness of evaluating the BvN decomposition density. This hybrid character — information-theoretic security for one entity, computational security from combinatorial hardness for another — is what distinguishes Combinatorial Privacy from the established paradigms. The paradigm is not limited to Birkhoff polytopes; any combinatorial structure with provably non-unique decompositions and hard density evaluation could potentially serve as the embedding space.

Summary of when to use each approach. PolyVeil is best suited for settings where the computation is a sum or linear aggregate, exact results are needed, communication bandwidth is constrained, PKI is unavailable or undesirable, and the semi-honest model is appropriate. MPC is preferred when the computation is complex or when malicious security is required. HE is preferred when no shuffler is available but PKI exists. DP is preferred when approximate answers suffice and no trust infrastructure (neither shuffler nor PKI) is available.

7.5 Communication Complexity

Theorem 7.1. *The total communication is $O(kn^2)$ for the full protocol and $O(k)$ for the compressed variant. The shuffle requires $O(k)$ communication.*

Proof. In the full protocol (Algorithm 1), each client t sends:

- The masked matrix $D_t \in \mathbb{R}^{2n\times 2n}$ to the server (or aggregator): $(2n)^2 = 4n^2$ real-valued entries.
- The scalar $\eta_t \in \mathbb{R}$ to the shuffler (or noise aggregator): 1 real number.

Summing over k clients: $k\cdot(4n^2+1) = O(kn^2)$ real numbers total from clients to server/aggregator, plus k real numbers from clients to shuffler. The shuffler sends k scalars to the server. Total: $O(kn^2) + O(k) = O(kn^2)$.

In the compressed variant (Algorithm 2), each client t sends:

- The scalar $f_t = \alpha^* s_t + \eta_t \in \mathbb{R}$ to the server: 1 real number.
- The scalar $\eta_t \in \mathbb{R}$ to the shuffler: 1 real number.

Summing over k clients: $k \cdot 2 = 2k$ real numbers total. The shuffler sends k scalars to the server. Total: $2k + k = 3k = O(k)$ real numbers, independent of n. □

7.6 Attack Resilience

7.6.1 The Carlini et al. Attack on InstaHide and Why It Does Not Apply

InstaHide [11] was broken by Carlini et al. [12]. We describe their attack precisely and explain why each step fails against PolyVeil.

In InstaHide, a training dataset of private images $\{x_1, \ldots, x_m\}$ is encoded for release. Each encoding $\tilde{x}_a$ combines a private image with randomly selected public images via $\tilde{x}_a = \sum_{i\in T_a} c_{a,i}\, \sigma_{a,i} \odot x_i$, where T_a is a random subset of images, $c_{a,i}$ are mixing coefficients, $\sigma_{a,i} \in \{+1, -1\}^d$ are random coordinate-wise sign vectors, and $\odot$ denotes the Hadamard (element-wise) product. A critical feature is that each private image appears in *multiple* encodings across the dataset.

The Carlini et al. attack has two stages. In the first stage (graph recovery), the attackers compute the coordinate-wise squared inner product $\sum_{d'} (\tilde{x}_a)^2_{d'} (\tilde{x}_b)^2_{d'}$ between all pairs (a, b) of encodings. The random signs σ are eliminated by the squaring, and when two encodings a and b share a common private image x_i (i.e., $i \in T_a \cap T_b$), this quantity is detectably larger than when they share no private image. This allows the attackers to reconstruct a bipartite graph between encodings and private images. In the second stage (image recovery), the graph structure determines which encodings correspond to which private images, and the mixing equations $\tilde{x}_a = \sum_i c_{a,i}\sigma_{a,i} \odot x_i$ become a solvable linear system.

This attack fails against PolyVeil for two specific structural reasons. First, PolyVeil generates exactly one masked matrix D_t per client per protocol execution. There is no dataset of multiple encodings sharing common private components, so the graph recovery stage has no correlations to detect. Even across independent protocol executions (with fresh random decoy permutations each time), the masked matrices are statistically independent, providing no cross-execution signal. Second, PolyVeil uses convex matrix combinations rather than element-wise sign flips. The Hadamard product structure of InstaHide is what makes the squared-inner-product trick work (squaring eliminates the random signs coordinate by coordinate). PolyVeil's convex combination of permutation matrices has no analogous coordinate-wise structure; the entries of D_t are sums of contributions from multiple permutations, and there is no operation that "cancels" the decoy components while preserving the private component.

7.7 Post-Quantum Security

In the two-layer protocol (Section 8), the server's security is information-theoretic and therefore immune to quantum attack. The aggregator's security against likelihood-based attacks involves the #P-hardness of mixed discriminants and permanent-related computations. No quantum algorithm is known to compute the permanent in polynomial time, and the problem is widely believed to be hard for quantum computers (it is #P-hard, and #P is believed to be outside BQP). However, the quantum hardness of the permanent is not proved, and full aggregator security (Conjecture 8.5) depends on open complexity-theoretic questions. The post-quantum status of PolyVeil's computational layer is therefore plausible but not certain.

7.8 Relationship to Ciphertext Indistinguishability

Standard ciphertext indistinguishability (IND-CPA) requires that no polynomial-time adversary can distinguish encryptions of two chosen plaintexts with non-negligible advantage. In the two-layer protocol, the server achieves a guarantee strictly stronger than IND-CPA: the server's view (F, H) is *identically distributed* (statistical distance zero) for any two inputs with the same aggregate, against all adversaries regardless of computational power. This is the information-theoretic analogue of IND-CPA, sometimes called perfect secrecy for the server entity.

The aggregator faces a weaker (computational) guarantee: it can compute the posterior distribution over M_t given D_t in principle, but the natural approach (likelihood evaluation) is #P-hard (Theorem 8.6). Whether all polynomial-time attacks are ruled out remains an open conjecture (Conjecture 8.5).

7.9 Potential Attack Vectors

Collusion. If c clients reveal their η_t values to the server outside the shuffle, the server can compute $s_t = (f_t - \eta_t)/\alpha^*$ for those clients, reducing the anonymity set from k to $k - c$.

Malicious server. A server that deviates from the protocol (e.g., using different extraction vectors for different clients) could extract additional linear functionals of D_t. Extending PolyVeil to the malicious model would require verifiable computation.

Repeated executions with correlated data. Each execution is independently secure with fresh randomness. However, if the same clients run the protocol on correlated bitstreams across rounds, the server may detect correlations in f_t values. Cross-round correlation attacks remain an open question.

Small k. The shuffle provides $1/k$ anonymity per client. For $k = 3$, this is weak. Strong privacy requires large k.

8 The Two-Layer PolyVeil Protocol

The de-shuffling attack (Theorem 5.1) shows that no protocol variant in which the server sees identity-linked f_t values *and* shuffled η_t values separately can be secure. The root cause is that the integrality constraint $s_t \in \{0, \ldots, n\}$ allows the server to uniquely identify the true shuffle permutation.

We now present a corrected protocol with a *two-layer* architecture in which no single entity can learn individual data. The design addresses the reviewer critique that any aggregation-only protocol (where the server sees only $\sum f_t$ and $\sum \eta_t$) is trivially secure and does not require the Birkhoff polytope. In our two-layer protocol, the Birkhoff encoding is *essential* for the security of the aggregation layer.

8.1 Architecture Overview

The protocol involves three types of entities:

- **Clients** $t \in [k]$: each holds private $\mathbf{b}_t \in \{0,1\}^n$.
- **Aggregator** $\mathcal{A}$: receives masked matrices D_t and computes the scalar aggregate $F = \sum_t \mathbf{w}^T D_t \mathbf{y}$, which it sends to the server. The aggregator does *not* receive η_t values.
- **Server** $\mathcal{S}$: receives F from the aggregator and $H = \sum_t \eta_t$ from a separate noise-aggregation channel, and computes $S = (F - H)/\alpha^*$.

The key design principle is *separation of information*: the aggregator sees D_t (which encodes M_t) but not η_t; a separate channel delivers $\sum \eta_t$ to the server without the aggregator's involvement.

Algorithm 3 Two-Layer PolyVeil

Public parameters: n, $\alpha^* \in (0,1)$, $\mathbf{w}$, $\mathbf{y}$.
Entities: Aggregator $\mathcal{A}$, noise aggregator $\mathcal{B}$, server $\mathcal{S}$.

Client-Side Computation
for each client $t \in [k]$ in parallel **do**
  Encode $M_t = M(\mathbf{b}_t)$. Draw $K_t \geq 2$ uniform random permutations $P_{t,i} \in S_{2n}$.
  Sample $\alpha_{t,1}, \ldots, \alpha_{t,K_t} > 0$ with $\sum_i \alpha_{t,i} = 1 - \alpha^*$.
  Compute $D_t = \alpha^* M_t + \sum_i \alpha_{t,i} P_{t,i}$ and $\eta_t = \sum_i \alpha_{t,i}(\mathbf{w}^T P_{t,i}\mathbf{y})$.
  Send D_t to aggregator $\mathcal{A}$. Send η_t to noise aggregator $\mathcal{B}$.
end for
Aggregation
$\mathcal{A}$ computes $F = \sum_{t=1}^k \mathbf{w}^T D_t \mathbf{y}$ and sends F to server $\mathcal{S}$.
$\mathcal{B}$ computes $H = \sum_{t=1}^k \eta_t$ and sends H to server $\mathcal{S}$.
$\mathcal{S}$ computes $S = (F - H)/\alpha^*$.

8.2 Correctness

Theorem 8.1 (Correctness)**.** *Algorithm 3 outputs $S = \sum_t s_t$ exactly.*

Proof. The aggregator computes:

$$F = \sum_{t=1}^{k} \mathbf{w}^T D_t \, \mathbf{y} = \sum_{t=1}^{k} \mathbf{w}^T \left(\alpha^* M_t + \sum_{i=1}^{K_t} \alpha_{t,i} P_{t,i} \right) \mathbf{y}$$
$$= \sum_{t=1}^{k} \left(\alpha^* (\mathbf{w}^T M_t \, \mathbf{y}) + \sum_{i=1}^{K_t} \alpha_{t,i} (\mathbf{w}^T P_{t,i} \, \mathbf{y}) \right) = \sum_{t=1}^{k} (\alpha^* s_t + \eta_t) = \alpha^* \sum_{t=1}^{k} s_t + \sum_{t=1}^{k} \eta_t = \alpha^* S + H,$$

where we used Lemma 3.9 ($\mathbf{w}^T M_t \mathbf{y} = s_t$) and the definition $\eta_t = \sum_i \alpha_{t,i}(\mathbf{w}^T P_{t,i}\mathbf{y})$. The noise aggregator computes $H = \sum_{t=1}^k \eta_t$ independently. The server receives F and H and computes:

$$\frac{F - H}{\alpha^*} = \frac{(\alpha^* S + H) - H}{\alpha^*} = \frac{\alpha^* S}{\alpha^*} = S.$$

□

8.3 Layer 1: Information-Theoretic Security of the Server

The server $\mathcal{S}$ receives only two scalars (F, H).

Theorem 8.2 (Perfect simulation-based security of the server)**.** *For any two input configurations $\mathcal{D}$ and $\mathcal{D}'$ with $\sum_t s_t = \sum_t s'_t = S$, the server's view (F, H) is identically distributed. There exists a simulator $\mathcal{S}^{\text{sim}}$ such that $\mathcal{S}^{\text{sim}}(S) \equiv (F, H)$ for all inputs with aggregate S.*

Proof. The proof proceeds in four steps: we characterize the noise distribution, establish its independence from inputs, derive the server's view distribution as a function of S alone, and construct the simulator.

The noise distribution μ is independent of inputs. Each $\eta_t = \sum_{i=1}^{K_t} \alpha_{t,i}(\mathbf{w}^T P_{t,i}\mathbf{y})$, where the permutation matrices $P_{t,i} \xleftarrow{\$} S_{2n}$ and the coefficients $(\alpha_{t,1}, \ldots, \alpha_{t,K_t})$ are drawn from the protocol's randomness distribution, independently of $\mathbf{b}_t$. Let μ denote the induced probability distribution (density) of a single η_t. This distribution is fully determined by the public parameters (n, α^*, K_t) and the coefficient distribution — it does not depend on $\mathbf{b}_t$ or s_t.

Formally: let $\Omega_t = S_{2n}^{K_t} \times \Delta_{K_t}$ be the probability space of client t's randomness, where Δ_{K_t} is the coefficient simplex. The map $\omega_t \mapsto \eta_t(\omega_t)$ is measurable and ω_t is sampled independently of $\mathbf{b}_t$. Therefore $\eta_t \perp\!\!\!\perp \mathbf{b}_t$ (statistical independence).

Since the randomness is also independent across clients, $\eta_1, \ldots, \eta_k$ are mutually independent, each with distribution μ:

$$(\eta_1, \ldots, \eta_k) \ \sim \ \mu^{\otimes k}. \tag{10}$$

Distribution of H. The noise aggregate is $H = \sum_{t=1}^k \eta_t$. Since $(\eta_1, \ldots, \eta_k) \sim \mu^{\otimes k}$ independently of $(\mathbf{b}_1, \ldots, \mathbf{b}_k)$, the distribution of H is the k-fold convolution μ^{*k}, which is independent of the input configuration $\mathcal{D}$. For any measurable set $B \subseteq \mathbb{R}$:

$$\Pr[H \in B \mid \mathcal{D}] = \Pr[H \in B] = \mu^{*k}(B), \qquad \forall\, \mathcal{D}. \tag{11}$$

F is a deterministic function of S and H. From the correctness proof (Theorem 8.1):

$$F = \sum_{t=1}^k \mathbf{w}^T D_t \mathbf{y} = \sum_{t=1}^k (\alpha^* s_t + \eta_t) = \alpha^* \sum_{t=1}^k s_t + \sum_{t=1}^k \eta_t = \alpha^* S + H.$$

Therefore F is a deterministic, invertible function of (S, H). The server's view is:

$$(F,\ H) = (\alpha^* S + H,\ \ H).$$

Distributional identity. For any measurable set $A \subseteq \mathbb{R}^2$ and any input configuration $\mathcal{D}$ with $\sum_t s_t = S$:

$$\begin{aligned}\Pr[(F, H) \in A \mid \mathcal{D}] &= \Pr[(\alpha^* S + H,\ \ H) \in A] \\ &= \int_{\mathbb{R}} \mathbf{1}\big[(\alpha^* S + h,\ \ h) \in A\big]\, d\mu^{*k}(h),\end{aligned} \tag{12}$$

where the second equality substitutes $F = \alpha^* S + H$ and uses (11) to integrate H against μ^{*k}.

For a different input $\mathcal{D}'$ with $\sum_t s'_t = S$:

$$\Pr\big[(F, H) \in A \mid \mathcal{D}'\big] = \int_{\mathbb{R}} \mathbf{1}\big[(\alpha^* S + h,\ \ h) \in A\big]\, d\mu^{*k}(h).$$

The two expressions are identical: same integrand (both use S), same measure (μ^{*k} is independent of $\mathcal{D}$). Therefore $\Pr[(F, H) \in A \mid \mathcal{D}] = \Pr[(F, H) \in A \mid \mathcal{D}']$ for all A, i.e., the distributions are identical.

Simulator construction. Define $\mathcal{S}^{\text{sim}}$ on input (S, k, n, α^*):

(a) For $t = 1, \ldots, k$: draw K_t independent uniform random permutations $\hat{P}_{t,i} \in S_{2n}$ and coefficients $\hat{\alpha}_{t,i}$ from the same distribution as the real protocol.

(b) Compute $\hat{\eta}_t = \sum_i \hat{\alpha}_{t,i}(\mathbf{w}^T \hat{P}_{t,i}\mathbf{y})$ for each t.

(c) Compute $\hat{H} = \sum_{t=1}^{k} \hat{\eta}_t$ and $\hat{F} = \alpha^* S + \hat{H}$.

(d) Output $(\hat{F},\ \hat{H})$.

By construction, $(\hat{\eta}_1, \ldots, \hat{\eta}_k) \sim \mu^{\otimes k}$, so $\hat{H} \sim \mu^{*k}$, and $(\hat{F}, \hat{H}) = (\alpha^* S + \hat{H},\ \hat{H})$. By (12), this has the same distribution as the real view for any $\mathcal{D}$ with aggregate S. □

This is information-theoretic security: statistical distance zero, holding against computationally unbounded adversaries. The proof uses only that η_t is independent of $\mathbf{b}_t$ — any noise generation process with this property would suffice for the server's security alone.

8.4 Layer 2: Computational Security of the Aggregator via #P-Hardness

The aggregator $\mathcal{A}$ sees the individual matrices $D_1, \ldots, D_k$ but does *not* receive η_t or f_t as separate scalars. To recover client t's data, the aggregator must extract M_t from $D_t = \alpha^* M_t + (1 - \alpha^*) R_t$. We now show that this task is computationally hard.

8.4.1 The Aggregator's Inference Problem

Given D_t, the aggregator can enumerate the set of *consistent* permutation matrices:

$$\mathcal{L}(D_t, \alpha^*) = \{M' \in S_{2n} : (D_t - \alpha^* M')/(1 - \alpha^*) \in \mathcal{B}_{2n}\}.$$

Under the interior condition ($\alpha^* \leq \min_{ij}(D_t)_{ij}$, Proposition 6.1), this set contains all $(2n)!$ permutation matrices. But consistency alone does not prevent inference: the aggregator can compute the *posterior probability* of each candidate:

$$\Pr(M_t = M' \mid D_t) \ = \ \frac{\nu(R')}{\sum_{M'' \in \mathcal{L}} \nu(R'')}, \tag{13}$$

where $R' = (D_t - \alpha^* M')/(1 - \alpha^*)$ and ν is the density of the random decoy component R_t over $\mathcal{B}_{2n}$.

Computing this posterior requires evaluating the density $\nu(R')$. We show this is #P-hard.

8.4.2 The Density of a Random Doubly Stochastic Matrix

Definition 8.3 (Random doubly stochastic matrix from random convex combination)**.** *Let $P_1, \ldots, P_K$ be i.i.d. uniform random permutation matrices in S_{2n}, and let $(\alpha_1, \ldots, \alpha_K)$ be positive coefficients with $\sum_i \alpha_i = 1 - \alpha^*$, drawn from some continuous distribution on the simplex. The random doubly stochastic matrix $R = \frac{1}{1-\alpha^*} \sum_i \alpha_i P_i$ has a distribution ν over $\mathcal{B}_{2n}$.*

The density $\nu(R')$ at a point $R' \in \mathcal{B}_{2n}$ is the probability (density) that the random process produces exactly R'. Computing this requires marginalizing over all possible permutation matrices and coefficient vectors that yield R':

$$\nu(R') = \sum_{\sigma_1,\ldots,\sigma_K \in S_{2n}} \int_{\Delta_K} \mathbf{1}\left[\frac{1}{1-\alpha^*} \sum_{i=1}^{K} \alpha_i P_{\sigma_i} = R'\right] \cdot \frac{1}{((2n)!)^K} \cdot g(\alpha_1, \ldots, \alpha_K)\, d\alpha, \tag{14}$$

where g is the density of the coefficient distribution on the K-simplex, and $\Delta_K = \{(\alpha_1, \ldots, \alpha_K) : \alpha_i > 0,\ \sum \alpha_i = 1 - \alpha^*\}$.

8.4.3 Reduction to the Permanent

We present evidence that the aggregator's inference problem is computationally hard, connecting it to the #P-completeness of the permanent. We state the strongest claim as a conjecture and prove a weaker but rigorous supporting result.

Proposition 8.4 (Evaluating ν requires counting perfect matchings)**.** *The summation in* (14) *ranges over all K-tuples of permutation matrices in* $\mathrm{Supp}(R') = \{Q \in S_{2n} : Q_{ij} = 1 \implies R'_{ij} > 0\}$. *The cardinality* $|\mathrm{Supp}(R')| = \mathrm{perm}(A(R'))$, *where* $A_{ij} = \mathbf{1}[R'_{ij} > 0]$ *and* perm *denotes the permanent. By Valiant [15], computing* $\mathrm{perm}(A)$ *for 0/1 matrices is #P-complete. The summation in* (14) *therefore has* $\mathrm{perm}(A(R'))^K$ *nonzero terms.*

Proof. We derive the support constraint as follows.

Nonzero indicator requires support containment. The indicator function in (14) is $\mathbf{1}[\frac{1}{1-\alpha^*} \sum_{i=1}^{K} \alpha_i P_{\sigma_i} = R']$. For this to be nonzero, we need $\sum_i \alpha_i P_{\sigma_i} = (1-\alpha^*)R'$ entry-wise. Consider any entry (a, b):

$$\sum_{i=1}^{K} \alpha_i (P_{\sigma_i})_{ab} = (1-\alpha^*) R'_{ab}.$$

Since each $\alpha_i > 0$ and each $(P_{\sigma_i})_{ab} \in \{0, 1\}$ (permutation matrices are 0/1 matrices), the left side is $\sum_{i:(P_{\sigma_i})_{ab}=1} \alpha_i \geq 0$, with equality iff $(P_{\sigma_i})_{ab} = 0$ for all $i \in [K]$.

If $R'_{ab} > 0$, then $(1-\alpha^*)R'_{ab} > 0$, so at least one P_{σ_i} must have $(P_{\sigma_i})_{ab} = 1$.

If $R'_{ab} = 0$, then $(1-\alpha^*)R'_{ab} = 0$, so $\sum_i \alpha_i (P_{\sigma_i})_{ab} = 0$, which (since $\alpha_i > 0$) forces $(P_{\sigma_i})_{ab} = 0$ for all i.

Each permutation must lie in the support. Combining both cases: for each $i \in [K]$ and each entry (a, b), if $(P_{\sigma_i})_{ab} = 1$ then we must have $R'_{ab} > 0$. Equivalently, each P_{σ_i} places its 1's only at positions where R' is positive:

$$(P_{\sigma_i})_{ab} = 1 \implies R'_{ab} > 0, \qquad \forall\, i \in [K],\ \forall\, (a, b).$$

This means $P_{\sigma_i} \in \mathrm{Supp}(R') := \{Q \in S_{2n} : Q_{ab} = 1 \implies R'_{ab} > 0\}$.

$|\mathrm{Supp}(R')|$ *equals the permanent.* Define the 0/1 support matrix $A = A(R') \in \{0,1\}^{2n\times 2n}$ by $A_{ab} = \mathbf{1}[R'_{ab} > 0]$. A permutation matrix Q belongs to $\mathrm{Supp}(R')$ if and only if $Q_{ab} = 1 \implies A_{ab} = 1$, i.e., Q is "contained in" A. The number of such permutation matrices equals the number of ways to choose one 1-entry per row and per column from positions where $A_{ab} = 1$. This is exactly the number of perfect matchings in the bipartite graph $G(A)$ with edge set $\{(a,b) : A_{ab} = 1\}$, which equals the permanent:

$$|\mathrm{Supp}(R')| = \sum_{\sigma\in S_{2n}} \prod_{a=1}^{2n} A_{a,\sigma(a)} = \mathrm{perm}(A(R')).$$

Number of nonzero terms in (14). The outer sum in (14) ranges over all K-tuples $(\sigma_1,\ldots,\sigma_K) \in S_{2n}^K$. By the support containment argument above, a tuple contributes a nonzero term only if each $P_{\sigma_i} \in \mathrm{Supp}(R')$. There are $|\mathrm{Supp}(R')|$ choices for each σ_i, and the choices are independent, so the number of nonzero tuples is $|\mathrm{Supp}(R')|^K = \mathrm{perm}(A(R'))^K$.

By Valiant [15], computing $\mathrm{perm}(A)$ for 0/1 matrices is #P-complete. Therefore the summation in (14) has $\mathrm{perm}(A(R'))^K$ nonzero terms, and even determining this count is #P-hard. □

Proposition 8.4 establishes that the *natural algorithm* for computing the posterior — evaluate $\nu(R')$ for each candidate M' using (14) — requires summing over a #P-hard number of terms. However, this does not rule out the existence of alternative algorithms that recover M_t without computing ν.

Conjecture 8.5 (Computational hardness of the aggregator's inference)**.** *No polynomial-time algorithm can, given D_t drawn from $\alpha^* M_t + (1-\alpha^*)\nu$ (where M_t is a uniformly random permutation matrix), recover M_t with probability non-negligibly better than $1/(2n)!$.*

We do not prove this conjecture in full generality. A complete proof would require showing that *any* efficient algorithm for recovering M_t can be transformed into an efficient algorithm for a #P-hard problem — a reduction in the opposite direction from what Proposition 8.4 establishes. Such a reduction faces two technical obstacles: (a) the permanent is #P-hard in the *worst case*, whereas the aggregator faces a *distributional* instance (the residual R was drawn from ν), and average-case hardness of the permanent is an open problem [17]; and (b) even worst-case hardness of evaluating ν does not rule out algorithms that bypass density evaluation entirely.

We do, however, prove the following supporting result.

Theorem 8.6 (Aggregator security against likelihood-based attacks)**.** *Any algorithm that recovers M_t from D_t by computing or approximating the likelihood ratio $\nu(R')/\nu(R'')$ for candidate permutation matrices M', M'' must solve an instance of the* mixed discriminant *problem, which is #P-hard (Barvinok, 1996 [16]).*

Proof. We show that computing $\nu(R')$ requires evaluating a quantity that reduces to the mixed discriminant, which is #P-hard.

Decomposing $\nu(R')$ into per-tuple integrals. From (14), the density at R' is:

$$\nu(R') = \sum_{(\sigma_1,\ldots,\sigma_K)\in \mathrm{Supp}(R')^K} \frac{1}{((2n)!)^K} \cdot I(\sigma_1,\ldots,\sigma_K;\ R'),$$

where (by Proposition 8.4) only tuples with each $P_{\sigma_i} \in \mathrm{Supp}(R')$ contribute, and the per-tuple integral is:

$$I(\sigma_1,\ldots,\sigma_K;\ R') = \int_{\Delta_K} \mathbf{1}\left[\sum_{i=1}^{K} \alpha_i P_{\sigma_i} = (1-\alpha^*)R'\right] \cdot g(\alpha_1,\ldots,\alpha_K)\, d\alpha. \tag{15}$$

The constraint defines a convex polytope. Fix permutation matrices $P_{\sigma_1}, \ldots, P_{\sigma_K} \in \mathrm{Supp}(R')$. The constraint $\sum_i \alpha_i P_{\sigma_i} = (1-\alpha^*)R'$ is a system of $(2n)^2$ linear equations in K unknowns $\alpha_1, \ldots, \alpha_K$. Since the matrices P_{σ_i} are 0/1 and R' is fixed, each equation has the form:

$$\sum_{i:(P_{\sigma_i})_{ab}=1} \alpha_i = (1-\alpha^*)R'_{ab}, \qquad \forall\,(a,b) \in [2n]^2.$$

The feasible set for α is the intersection of the simplex $\Delta_K = \{\alpha \in \mathbb{R}^K_{>0} : \sum_i \alpha_i = 1-\alpha^*\}$ with the affine subspace defined by these equations. This intersection is a convex polytope $\mathcal{P}(\sigma_1, \ldots, \sigma_K; R') \subseteq \mathbb{R}^K$.

The integral equals a polytope volume. When the coefficient distribution g is uniform over the simplex (i.e., g is the Dirichlet$(1, \ldots, 1)$ density, proportional to a constant on Δ_K), the integral (15) equals:

$$I(\sigma_1, \ldots, \sigma_K; R') = C_K \cdot \mathrm{vol}_{d_0}\big(\mathcal{P}(\sigma_1, \ldots, \sigma_K; R')\big)\,,$$

where C_K is the normalizing constant of the Dirichlet distribution and $d_0 = \dim(\mathcal{P})$.

Connection to mixed discriminants. The total density is:

$$\nu(R') = \frac{C_K}{((2n)!)^K} \sum_{(\sigma_1,\ldots,\sigma_K)\in \mathrm{Supp}(R')^K} \mathrm{vol}_{d_0}\big(\mathcal{P}(\sigma_1, \ldots, \sigma_K; R')\big)\,. \tag{16}$$

Each polytope $\mathcal{P}(\sigma_1, \ldots, \sigma_K; R')$ is defined by the intersection of the simplex with a linear subspace determined by the permutation matrices $P_{\sigma_1}, \ldots, P_{\sigma_K}$.

Now recall the definition of the mixed discriminant. Given d positive semidefinite matrices $X_1, \ldots, X_d$ of size $d \times d$, the mixed discriminant is:

$$D(X_1, \ldots, X_d) = \frac{\partial^d}{\partial\lambda_1 \cdots \partial\lambda_d} \det\left(\sum_{j=1}^{d} \lambda_j X_j\right)\Bigg|_{\lambda_1=\cdots=\lambda_d=0}.$$

For the specific case $K = 2n$ with permutation matrices $P_{\sigma_1}, \ldots, P_{\sigma_{2n}}$, each P_{σ_i} can be viewed as a rank-1 matrix (in each row/column) and the constraint $\sum_i \alpha_i P_{\sigma_i} = (1-\alpha^*)R'$ involves mixed volumes of zonotopes defined by the permutations. Barvinok [16] showed that computing such mixed volumes (equivalently, mixed discriminants of matrices constructed from 0/1 permutations) is #P-hard.

The reduction. Computing the likelihood ratio $\nu(R')/\nu(R'')$ requires evaluating both $\nu(R')$ and $\nu(R'')$ (or their ratio). By (16), each evaluation involves summing $\mathrm{perm}(A(R'))^K$ polytope volumes, where each volume is a mixed discriminant. Since computing the mixed discriminant is #P-hard, and the sum involves a #P-hard number of such terms, any algorithm that evaluates $\nu(R')$ (even approximately, to within a polynomial factor) must solve a #P-hard problem.

Therefore, any algorithm that ranks candidates M' by likelihood (MAP estimation, likelihood ratio tests, or any Bayesian posterior computation) must evaluate ν and hence solve a #P-hard problem. □

Remark 8.7 (Significance and limitations)**.** *Theorem 8.6 proves that the* most natural class of attacks *— those based on Bayesian posterior computation — is computationally intractable. This eliminates the MAP estimator and any algorithm that ranks candidates by likelihood. It does* not *rule out algorithms that exploit structural properties of D_t (e.g., spectral methods, linear programming relaxations) to recover M_t without computing likelihoods.*

We contrast this with the situation for other noise distributions. If the decoy component were additive Gaussian noise ($D_t = \alpha^* M_t + \varepsilon$, $\varepsilon \sim N(0, \sigma^2 I)$), the likelihood $\Pr(D_t \mid M')$ would be a simple Gaussian density, computable in $O(n^2)$ time. The MAP estimate would equal the nearest permutation matrix to D_t/α^* in Frobenius norm, solvable by the Hungarian algorithm in $O(n^3)$. The Birkhoff encoding is therefore strictly harder to invert than Gaussian noise, because the likelihood involves a #P-hard summation that Gaussian noise avoids.

8.5 Formal Two-Layer Security Statement

Definition 8.8 (Two-layer security)**.** *A protocol has* two-layer security *against semi-honest adversaries if:*

(i) ***Information-theoretic server security:*** *the server's view is identically distributed for any two inputs with the same aggregate (statistical distance zero).*

(ii) ***Computational aggregator security against likelihood attacks:*** *no polynomial-time aggregator can compute or approximate the posterior distribution* (13) *over candidate permutation matrices, because doing so requires solving a #P-hard problem.*

(iii) ***Non-collusion:*** *the server and aggregator do not share their views.*

Theorem 8.9 (Two-Layer PolyVeil security — proved components)**.** *Under the non-collusion assumption, Algorithm 3 achieves:*

- *condition (i) by Theorem 8.2 (perfect simulation, unconditional);*
- *condition (ii) by Theorem 8.6 (likelihood-based attacks require solving #P-hard mixed discriminant), under the interior condition of Proposition 6.1.*

Remark 8.10 (What remains open)**.** *Full aggregator security (Conjecture 8.5) — ruling out* all *polynomial-time attacks, not just likelihood-based ones — remains an open problem. A proof would require either (a) a reduction from a #P-complete or NP-hard problem to the search problem "recover M_t from D_t," or (b) an average-case hardness result for the permanent over the specific distribution induced by ν. Both are significant open problems in computational complexity. The average-case hardness of the permanent has been studied by Lipton [17] and others, with partial results (e.g., hardness over finite fields) but no complete resolution over the reals.*

8.6 Why the Birkhoff Polytope Is Essential

We now explain precisely where the Birkhoff polytope machinery contributes and where it does not.

What the Birkhoff polytope does NOT provide. The server's information-theoretic security (Theorem 8.2) does not depend on the Birkhoff polytope. Any method of generating noise η_t independently of $\mathbf{b}_t$ would achieve the same result. In particular, the trivial protocol "each client sends s_t + noise to a trusted aggregator" achieves the same server security. This observation, correctly made in [4] and subsequent work on secure aggregation, is not the contribution of this paper.

What the Birkhoff polytope DOES provide. The aggregator's computational security against likelihood-based attacks (Theorem 8.6) *requires* the Birkhoff polytope. The aggregator sees $D_t = \alpha^* M_t + (1-\alpha^*)R_t$ and must extract M_t. The difficulty of this extraction is a consequence of three properties of the Birkhoff polytope, all of which are necessary:

1. **Super-exponentially many vertices.** $\mathcal{B}_{2n}$ has $(2n)!$ vertices (permutation matrices). This means the number of candidate explanations grows faster than $2^{cn \log n}$, providing a large search space.

2. **Non-uniqueness of decomposition.** Every interior point of $\mathcal{B}_{2n}$ admits multiple BvN decompositions. Under the interior condition, every permutation matrix is a *valid* component of some decomposition of D_t. This means the set of consistent candidates is as large as possible.

3. **#P-hardness of the likelihood.** Computing the posterior probability of each candidate M' requires evaluating $\nu(R')$, which involves a summation over $\mathrm{perm}(A(R'))^K$ terms (Proposition 8.4) and mixed discriminant computations that are #P-hard (Theorem 8.6). This is the crucial ingredient: it is not enough that many decompositions *exist* — the *likelihood-based approach* to determining which one was used must be computationally intractable. Whether non-likelihood-based approaches are also hard remains an open problem (Conjecture 8.5).

These three properties together constitute what we call *Combinatorial Privacy*: privacy derived from the hardness of inverting a combinatorial embedding. The paradigm is distinct from noise-based privacy (DP), hardness-based encryption (MPC, HE using factoring/LWE), and secret-sharing-based protocols.

9 Conclusion

We have presented PolyVeil and introduced Combinatorial Privacy as a paradigm for privacy-preserving aggregation.

Our analysis proceeded in three stages. First, we described the basic protocol for private Boolean sums using Birkhoff polytope encoding and proved its correctness. Second, we identified a fatal vulnerability in the naive protocol — the de-shuffling attack (Theorem 5.1), which allows a semi-honest server to recover all individual data with probability 1 by exploiting the integrality constraint on bit counts.

Third, we developed the Two-Layer PolyVeil protocol (Algorithm 3), which achieves provable security through a separation-of-information architecture. The main server receives only aggregate scalars and achieves *perfect simulation-based security* (Theorem 8.2): its view is identically distributed for any two inputs with the same aggregate, against all adversaries regardless of computational power. A separate aggregator receives Birkhoff-encoded matrices but faces a computational barrier: we proved that likelihood-based inference requires solving #P-hard problems (Theorem 8.6), specifically the mixed discriminant and permanent. Whether all polynomial-time attacks (not just likelihood-based ones) are ruled out remains an open conjecture (Conjecture 8.5).

The Birkhoff polytope plays no role in the server's information-theoretic security (any aggregation protocol achieves that), but it is *essential* for the aggregator's computational barrier. Replacing the Birkhoff encoding with Gaussian noise would make the aggregator's inference trivially easy (Remark 8.7). This specificity — computational hardness from the combinatorial structure of a polytope's decompositions — is what distinguishes Combinatorial Privacy from both noise-based (DP) and number-theoretic (MPC, HE) approaches.

Future directions include formalizing average-case hardness of the aggregator's inference problem, extending to the malicious model, and investigating whether non-likelihood attacks can also be ruled out.

A Background on Simulation-Based Security Proofs

This appendix provides a self-contained introduction to the simulation paradigm for proving security of cryptographic protocols. It is a prerequisite for understanding the security proofs in Section 5.

A.1 The Problem That Simulation Solves

Consider a protocol Π where k parties hold private inputs $x_1, \ldots, x_k$ and jointly compute a function $f(x_1, \ldots, x_k) = y$. During execution, each party sends and receives messages. The *view* of party i, denoted $\text{View}_i^{\Pi}(x_1, \ldots, x_k)$, is the random variable consisting of party i's input x_i, its random coins r_i, and the sequence of all messages $m_1, m_2, \ldots$ it receives during execution. The fundamental question is whether View_i reveals information about other parties' inputs beyond what is already implied by x_i and y.

Intuitively, a protocol is "secure" if View_i is "no more informative" than (x_i, y). The simulation paradigm formalizes this by requiring the existence of an algorithm that can *fabricate* a fake view, using only (x_i, y) as input, such that the fake view is distributed identically to the real view.

A.2 Formal Definition

Definition A.1 (Simulation-based security, semi-honest model)**.** *A k-party protocol* Π securely computes f *in the semi-honest model if for each party $i \in [k]$, there exists a probabilistic polynomial-time algorithm $\mathcal{S}_i$ (the* simulator *for party i) such that for all input vectors $(x_1, \ldots, x_k)$ in the domain of f:*

$$\{\mathcal{S}_i(x_i,\ f(x_1, \ldots, x_k))\} \ \equiv\ \left\{\text{View}_i^{\Pi}(x_1, \ldots, x_k)\right\},$$

where $\equiv$ denotes either identical distributions (information-theoretic, or perfect *security) or computational indistinguishability (computational security). The left-hand side is the simulator's output distribution; the right-hand side is the real view's distribution.*

The simulator $\mathcal{S}_i$ receives *only* what party i is "supposed to know" after the protocol ends: its own input x_i and the output $y = f(x_1, \ldots, x_k)$. It does *not* receive any other party's input, the random coins of other parties, or the messages exchanged during the real protocol. Despite this limited input, the simulator must produce a fake transcript whose distribution matches the real one perfectly (or computationally indistinguishably).

A.3 How to Construct a Simulation Proof

A simulation proof proceeds in three stages.

In the first stage, one precisely defines the real view. For PolyVeil, the server's real view consists of the doubly stochastic matrices $D_1, \ldots, D_k$ (each linked to a client identity) and the shuffled noise sequence $(\eta_{\pi(1)}, \ldots, \eta_{\pi(k)})$. These are random variables whose joint distribution depends on all clients' inputs $\mathbf{b}_1, \ldots, \mathbf{b}_k$ and the random coins (decoy permutations, coefficients, shuffle permutation).

In the second stage, one constructs the simulator $\mathcal{S}$. The simulator receives only the aggregate $S = \sum_t s_t$ and the public parameters (k, n, α^*). It must produce a fake tuple $(D'_1, \ldots, D'_k, \tilde{\eta}'_1, \ldots, \tilde{\eta}'_k)$ with the same distribution as the real view. The typical construction is: choose fictitious inputs $\mathbf{b}'_t$ with $\sum s'_t = S$, run the real protocol honestly with these fictitious inputs and fresh randomness, and output the resulting messages.

In the third stage, one proves that the simulator's output is distributed identically to the real view. This is usually the hardest step. It requires showing that the joint distribution of all

messages is the same regardless of which specific inputs (with the same aggregate S) were used. The proof typically identifies which components of the view depend on the private inputs and which do not, and shows that the input-dependent components are "masked" by the input-independent randomness.

A.4 Why Simulation Implies Security

Proposition A.2. *If a simulator $\mathcal{S}_i$ with $\mathcal{S}_i(x_i, y) \equiv \mathrm{View}_i^{\Pi}(x_1, \ldots, x_k)$ exists, then for any function g (representing any "information extraction" strategy):*

$$g\big(\mathrm{View}_i^{\Pi}(x_1, \ldots, x_k)\big) \;\equiv\; g(\mathcal{S}_i(x_i, y))\,.$$

Proof. Let $X = \mathrm{View}_i^{\Pi}(x_1, \ldots, x_k)$ and $Y = \mathcal{S}_i(x_i, y)$. By assumption, $X \equiv Y$ (identical distributions). We need to show $g(X) \equiv g(Y)$ for any measurable g.

For any measurable set B:

$$\Pr[g(X) \in B] = \Pr[X \in g^{-1}(B)] = \Pr[Y \in g^{-1}(B)] = \Pr[g(Y) \in B],$$

where the second equality uses $X \equiv Y$ (applied to the measurable set $g^{-1}(B)$). Since $\Pr[g(X) \in B] = \Pr[g(Y) \in B]$ for all measurable B, the distributions of $g(X)$ and $g(Y)$ are identical. (This is the pushforward property: if $\mu_X = \mu_Y$ then $g_*\mu_X = g_*\mu_Y$.) □

The consequence is that any information $g(\mathrm{View}_i)$ that the adversary extracts from the real view could equally well have been extracted from (x_i, y) alone (via $g(\mathcal{S}_i(x_i, y))$, which is a function only of x_i and y). Therefore the protocol reveals no information beyond what (x_i, y) already implies.

A.5 Application to PolyVeil

In PolyVeil, the "adversary" is the honest-but-curious server. The server has no private input of its own ($x_i = \emptyset$). The function being computed is $f(\mathbf{b}_1, \ldots, \mathbf{b}_k) = S = \sum_t s_t$. Therefore the simulator receives only (S, k, n, α^*).

In the basic PolyVeil protocol (Algorithms 1 and 2), the server sees identity-linked values that allow deterministic de-shuffling (Theorem 5.1), so the basic protocol does not achieve simulation-based security. The Two-Layer PolyVeil protocol (Algorithm 3) corrects this: the server receives only two aggregate scalars (F, H), and its view is perfectly simulable from the aggregate S alone (Theorem 8.2). The aggregator, which sees individual Birkhoff-encoded matrices D_t but not the noise values η_t, faces a computational barrier: recovering M_t from D_t requires evaluating a density that is #P-hard to compute (Theorem 8.6).

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